Parabola

DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

Parabola

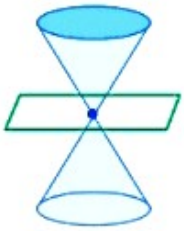
CONIC SECTION

When a cone (or double cone) is cut by a plane, various figures like point, pair of straight lines, circle, parabola, hyperbola or ellipse can be obtained. That's why, these figures are collectively called CONIC SECTION.

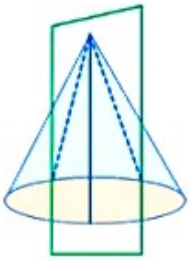
We can observe that we are surrounded by various objects which have the shape of conic sections. Hence it has become very important to study and explore these conics.

Section of right circular cone by various planes :

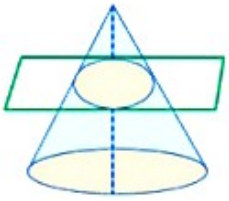
When we cut a double right circular cone by a plane parallel to base at the common vertex, what we obtain on the plane is a point.



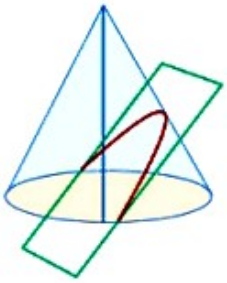
When we cut a right circular cone by any plane through its vertex, we obtain a pair of straight lines passing through its vertex.



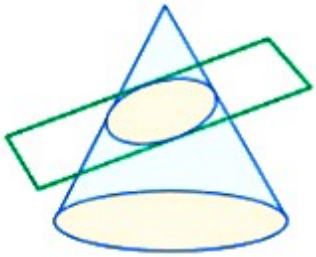
When we cut a right circular cone by a plane parallel to its base, we obtain a circle as shown in the figure.



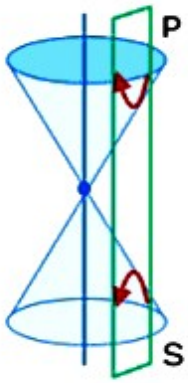
When we cut a right circular cone by a plane parallel to a generator of cone, we obtain a parabola.



When we cut a right circular cone by a plane which is neither parallel to any generator/axis nor parallel to base, the conic obtained is an ellipse.

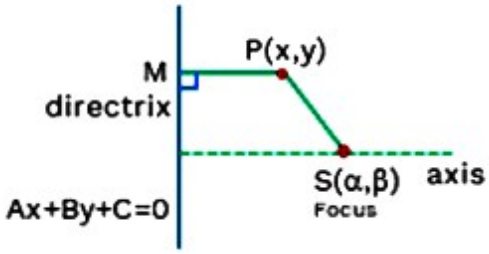


When we cut a double right circular cone by plane, parallel to its common axis, we obtain a hyperbola.



Hence, we can say that whether it's a point or a pair of intersecting straight lines or a circle or a parabola or ellipse and hyperbola, they are all conic.

The conic section is defined as the locus of a point which moves in such a way that the ratio of its distance from a given fixed point, which is called focus, to perpendicular distance from a given fixed straight line, which is called directrix, is always constant and equal to 'e'.



Here 'e' is called eccentricity of a conic i.e.,

$$\frac{PS}{PM} = e$$

A line passing through focus and perpendicular to directrix is called axis of the conic. The vertex of conic is the point where the curve intersects its axis.

$$\frac{PS}{PM} = e \Rightarrow PS^2 = e^2 PM^2$$

$$\Rightarrow (x-\alpha)^2 + (y-\beta)^2 = e^2 \left(\frac{Ax + By + C}{\sqrt{A^2 + B^2}} \right)^2$$

Simplifying it, we get an equation of the form:
$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$

Distinguishing various conics :

The nature of the conic section is decided based on two things:

- The position of the focus S with respect to the directrix of conic.
- The value of the eccentricity 'e'.

Now, we get two different cases:

Case-I :

When The Focus Lies On The Directrix (De-generated conic):

In this case $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

Here, the general equation of a conic represents a pair of straight lines if

- $e > 1 \Rightarrow h^2 > ab$, the reslulting lines will be real & distinct, intersecting at focus S.
- $e = 0 \Rightarrow h^2 = ab$, here the lines will be coincident.
- $e < 1 \Rightarrow h^2 < ab$, in this case, the lines will be imaginary.

Case-II :

When The Focus Does Not Lie on the Directrix (Non de-generated conic) :

In this case $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$ and conic represents:

A parabola	An ellipse	A hyperbola	Rectangular Hyperbola	Circle
$e = 1$	$0 < e < 1$	$e > 1$	$e = \sqrt{2}$	$e = 0$
$h^2 = ab$	$h^2 < ab$	$h^2 > ab$	$h^2 > ab$; $a + b = 0$	$h = 0, a = b$

Note :

- For pair of straight lines $e \to \infty$
- All second degree terms, in case of parabola, always form a perfect square.

Basic terms of a Conic :

- **Focus**: It is the fixed point in the definition of conic.
- **Directrix:** It is the fixed line present in the definition of the conic.
- **Axis:** It is the line passing through the focus and is perpendicular to the directrix of the conic.
- **Vertex:** It is represented by the points of intersection of the conic and it's axis.
- **Centre**: It is the point which bisects every chord of the conic passing through it.
- **Latus-rectum**: It is the chord passing through the focus and perpendicular to the axis of conic.
- **Double ordinate:** It is the chord which is perpendicular to the axis of parabola or parallel to its directrix.

Q. **What conic does $13x^2 - 18xy + 37y^2 + 2x + 14y - 2 = 0$ represent?**

Sol. Compare the given equation with
$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$\therefore$ $a = 13, h = -9, b = 37, g = 1, f = 7, c = -2$

Then $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$= (13)(37)(-2) + 2(7)(1)(-9) - 13(7)^2 - 37(1)^2 + 2(-9)^2$

$= -962 - 126 - 637 - 37 + 162$

And also $h^2 = (-9)^2 = 81$ and $ab = 13 \times 37 = 481$

Here $h^2 - ab < 0$

So we have $h^2 - ab < 0$ and Ä$\neq 0$..

Hence the given equation represents an ellipse.

Q. **For what value of λ the equation of conic $2xy + 4x - 6y + \lambda = 0$ represents two real intersecting straight lines? if $\lambda = 17$ then this equation represents ?**

Sol. Comparing the given equation of conic with
$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$\therefore$ $a = 0, b = 0, h = 1, g = 2, f = -3, c = \lambda$

For two intersecting real lines

$h^2 - ab \geq 0$ and $\Delta = 0$

here $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

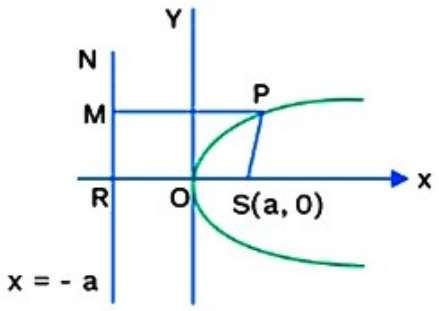
$= 0 - 12 - 0 - 0 - \lambda(1)^2$

$= -12 - \lambda = 0$

$\therefore \lambda = -12$
and $h^2 - ab = 1$
hence for $\lambda = -12$ above equation always represent real intersecting lines.
If $\lambda = 17$ then $\Delta \neq 0$ and $h^2 - ab > 0$
So we have $\Delta \neq 0$ and $h^2 - ab > 0$.
Hence the given equation represents a Hyperbola.

PARABOLA

If a point moves in a plane in such a way that its distance from a fixed point, i.e. focus, is equal to its perpendicular distance from a fixed straight line, i.e. directrix, then the locus of this point is known as a parabola. Eccentricity of parabola is unity, i.e. 1.



Standard Equation of a parabola:

Let S be the focus and RN be the directrix of a parabola. From S, SR is drawn perpendicular to the directrix RN.

Suppose, O is the middle point of SR. Now, O is taken as the origin and OS is taken as x-axis and OY which is perpendicular to OS is taken as the y -axis.

Let SR = 2a , then OR = OS = a

Now, S ≡ (a, 0) and the equation of RN is x = – a or x + a = 0

Now, suppose P(x, y) be any point lying on the parabola.

By the definition of parabola, PS = PM.

$$\Rightarrow \sqrt{(x-a)^2+(y-0)^2} = \frac{|x+a|}{\sqrt{1^2+0}}$$

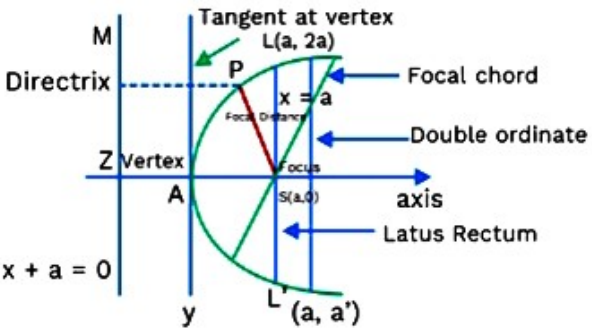
$$\Rightarrow \sqrt{(x-a)^2+y^2} = |x+a|$$

or $(x-a)^2 + y^2 = (x+a)^2$

or $x^2 - 2ax + a^2 + y^2 = x^2 + 2xa + a^2$

or $y^2 = 4ax$

which is the required equation.



Terms related to Parabola

(1) Axis

It is the straight line which passes through the focus and is perpendicular to the directrix. For the parabola $y^2 = 4ax$, x-axis is the axis. Since equation has even power of y therefore the parabola is symmetric about x-axis i.e., about its axis.

(2) Vertex

The point of connection of a parabola and its axis is known as the Vertex of the Parabola. For example, in the parabola $y^2 = 4ax$, O(0, 0) is the vertex.

(3) Focal Distance :

It is known by the distance of any point P(x, y) on the parabola from it's focus.
The focal distance of P is also equal to the perpendicular distance of the point P from the directrix

(4) Double Ordinate :

It is the chord drawn perpendicular to the axis of Parabola or parallel to Directrix.

(5) Focal Chord:

Any chord of the parabola which passes through the focus can be called Focal Chord.

(6) Latus Rectum :

A latus rectum is simply a double ordinate which passes through the focus of parabola. The extremities of the latus rectum are L(a, 2a) and L'(a, –2a). Thus, length of the latus rectum LL' is equal to 4a.

(7) Parametric Equation of Parabola :

In the parabola $y^2 = 4ax$, if we put $x = at^2$ and $y = 2at$, it completely satisfies the given equation. So, we can represent any point on this parabola by $(at^2, 2at)$ which is also called as 't' point.

Point to Remember!!!

(i) The length of the latus rectum is 2 times the perpendicular distance of focus from the directrix.

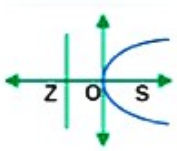
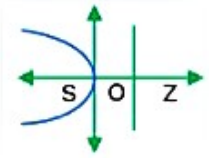
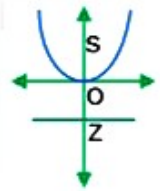
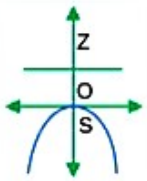
(ii) If $y^2 = kx$, then length of the latus rectum = k.

(iii) If any two parabolas have the same latus rectum, then both are considered to be equal.

(iv) Both ends of a double ordinate of a parabola are represented by $(at^2, 2at)$ and $(at^2, -2at)$.

(v) A parabola does not have any centre, unlike circle, ellipse or hyperbola all of which have centre.

Other Standard parabola :

Equation of Parabola	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
(a) Graphs				
(b) Eccentricity	e = 1	e = 1	e = 1	e = 1
(c) focus	S(a, 0)	S(-a, 0)	S(0, a)	S(0, - a)
(d) equation of directrix	x + a = 0	x − a =0	y + a = 0	y − a = 0
(e) equation of axis	y = 0	y = 0	x = 0	x = 0
(f) Vertex	O(0, 0)	O(0, 0)	O(0, 0)	O(0, 0)
(g) Extremities of latus rectum	(a, ± 2a)	(-a, ± 2a)	(±2a, a)	(±2a, − a)
(h) length of latus rectum	4a	4a	4a	4a
(i) equation of tangent of vertex	x = 0	x = 0	y = 0	y = 0
(j) parametric coordinates of any point on parabola	$P(at^2, 2at)$	$P(-at^2, 2at)$	$P(2at, at^2)$	$P(2at, -at^2)$

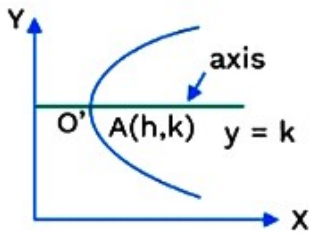
DEDUCTION TO GENERALIZED EQUATION OF PARABOLA

If the equation of a parabola is either in the form $x = \ell y^2 + my + n$ or $y = \ell x^2 + mx + n$

then it can be reduced into generalised form. For this we convert the given equation into the following forms

$(y-k)^2 = 4a(x-h)$ or $(x-h)^2 = 4a(y-k)$

And then we compare from the standard equation of parabola to find all its parameters.



(A) When the equation of parabola is :

$(y-k)^2 = 4a(x-h)$ ...(i)

Equation (i) is of the form $Y^2 = 4aX$

where $Y = y - k$ and $X = x - h$

(1) Axis of parabola is $Y = 0$, i.e., $y - k = 0$

$\Rightarrow y = k$

(2) Coordinates of vertex of parabola are given by

$X = 0$ and $Y = 0$

i.e. $x - h = 0$ and $y - k = 0$

$\therefore$ Vertex is (h, k)

(3) Tangent at the vertex to parabola (i) is given by

$X = 0$, i.e., $x - h = 0$

Therefore, tangent at the vertex is $x = h$.

(4) Coordinates of focus of parabola are given by

$X = a$ and $Y = 0$

i.e. by $x - h = a$ and $y - k = 0$

$\therefore$ Focus of parabola is $(a + h, k)$

(5) Equation of directrix of parabola is

$X = -a$

i. e. $x - h = -a$

Therefore, directrix of parabola is $x = h - a$.

(6) Length of latus rectum of parabola is $|4a|$.

(7) Coordinates of ends of latus rectum of parabola are given by

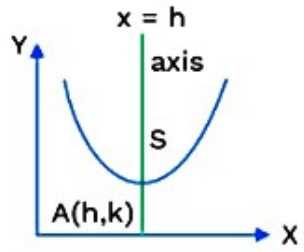
$X = a$ & $Y = \pm 2a$

i.e., by $x - h = a$, $-k = \pm 2a$ i.e. coordinate of latus rectum is $(a + h, k \pm 2a)$.

(8) Parametric equation is $x = h + at^2$ and $y = k + 2at$.

(B) When the equation of parabola is :

$(x - h)^2 = 4a(y - k)$ ... (i)



Equation (i) is of the form $X^2 = 4aY$

where $X = x - h$ and $Y = y - k$

(1) Axis of parabola is $X = 0$, i.e., $x - h = 0$

(2) Coordinates of vertex of parabola is given by

	$X = 0$	and	$Y = 0$
i.e. by	$x - h = 0$	and	$y - k = 0$
$\therefore$	$x = h$	and	$y = k$

(3) Equation of tangent at the vertex to parabola is 0

$Y = 0$ i.e., $y - k = 0$ or $y = k$

(4) Coordinates of focus of parabola are given by

	$X = 0$	and	$Y = a$
i.e., by	$x - h = 0$	and	$y - k = a$

$\therefore$ Focus of parabola is $(h, k + a)$..

(5) Equation of directrix of parabola (i) is given by

$Y = -a$ or $y - k = -a$ or $y = k - a$

(6) Length of latus rectum of parabola is $|4a|$.

(7) Coordinates of ends of latus rectum of parabola are given by

$Y = a$, $X = \pm 2a$

i.e., $y - k = a$, $x - h = \pm 2a$

$\therefore$ Ends of latus rectum are $(h \pm 2a, k + a)$

(8) Parametric equation is $x = h + 2$ at and $y = k + at^2$.

Equation of parabola	Vertex	Axis	Focus	Directrix	Parametric equation
$(y-k)^2 = 4a(x-h)$	(h, k)	$y = k$	$(h + a, k)$	$x + a - h = 0$	$x = h + at^2$, $y = k + 2at$
$(x-h)^2 = 4a(y-k)$	(h, k)	$x = h$	$(h, a + k)$	$y + a - k = 0$	$x = h + 2at$, $y = k + at^2$

Know the facts

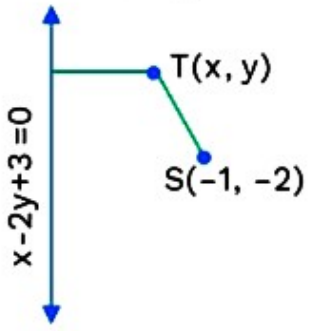
(i) For the parabola $x = Ay^2 + By + C$, the length of latus rectum is $\frac{1}{|A|}$ and axis is parallel to y-axis. If A is positive then parabola is concave up, if A is negative then it is concave down parabola.

(ii) For the parabola $x = Ay^2 + By + C$, the length of latus rectum is $\frac{1}{|A|}$ and axis of this parabola is parallel to x-axis.

If A is positive, then it opens towards the right and if A is negative then it is opening left parabola.

Q. **Determine the equation of the parabola whose focus is at (–1,–2) and the directrix is the line x – 2y + 3 = 0**

Sol. Let T(x, y) be a point on the parabola with focus is S(– 1, – 2) and the directrix x – 2y + 3 = 0. Draw PM perpendicular to directrix x – 2y + 3 = 0. Then by definition,



$ST = TM$

$\Rightarrow ST^2 = TM^2$

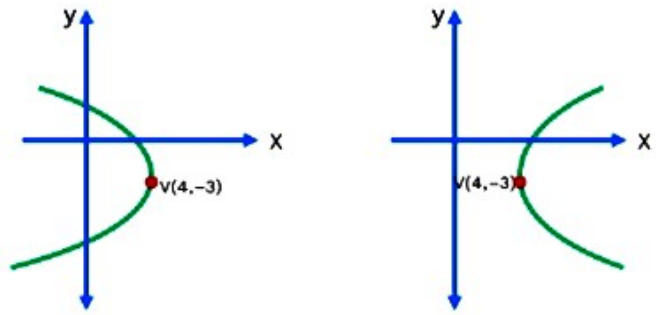
$$\Rightarrow (x+1)^2 + (y+2)^2 = \left(\frac{x-2y+3}{\sqrt{1+4}}\right)^2$$

$$\Rightarrow 5\left[(x+1)^2 + (y+2)^2\right] = (x-2y+3)^2$$

$$\Rightarrow 5\left(x^2 + y^2 + 2x + 4y + 5\right) = \left(x^2 + 4y^2 + 9 - 4xy + 6x - 12y\right)$$

$\Rightarrow 4x^2 + y^2 + 4xy + 4x + 32y + 16 = 0$
This is the equation of the needed parabola.

Q. **Find equation of parabola whose vertex is (4, – 3), latus rectum is 4 and axis is parallel to the x-axis.**



For 1st case

$(y + 3)^2 = -4a(x - 4)$
$(y + 3)^2 = -4(x - 4)$ $[4a = 4]$

For 2nd case

equation is $(y + 3)^2 = 4a(x - 4)$
$4a = 4$
hence $(y + 3)^2 = 4(x - 4)$

Q. **Find the equations of the directrix, axis, tangent at the vertex and latus rectum and the coordinates of the vertex and focus. Also find the parametric equations and the length of latus rectum for each of the following parabolas.**
(A) $y^2 - 12x - 8y + 4 = 0$ **(B) $4x^2 + 12x - 16y + 41 = 0$**

Sol. **Steps:** First we convert the second degree terms to perfect square. The equation will then reduce to either of two forms
$(y - k)^2 = 4a(x - h)$ or $(x - h)^2 = 4a(y - k)$
Now shift the origin to the point (h, k) to get either of two standard forms $y^2 = 4ax$ or $x^2 = 4ay$
We can write all propositions for the standard form and then for the given equation by transforming the origin to original positions.

(A) The given equation is
$y^2 - 12x - 8y + 4 = 0$ ... (1)
$y^2 - 8y + 16 - 12x - 12 = 0$
$\Rightarrow (y - 4)^2 = 12(x + 1)$ ... (2)

Shifting the origin to the point (–1, 4), and using transformation equations $x = X - 1$ and $y = Y + 4$, we get from (2),

$Y^2 = 12X$ ... (3)

Comparing with standard equation $y^2 = 4ax$, , we have

$4a = 12 \Rightarrow a = 3$

Now we will write all requirements for the equation (3) and then return the origin to original position by retransformation equations $X = x + 1$ and $Y = y - 4$.

(i) Equation of the Directrix: Equation of the directrix of the parabola (3) is

$X = -3 \Rightarrow x + 1 = -3 \Rightarrow x + 4 = 0$

So, the equation of the directrix of the given parabola (1) is $x + 4 = 0$.

(ii) Equation of the Axis : Equation of the axis of the parabola (3) is

$Y = 0 \Rightarrow y - 4 = 0$

So, the equation of the axis of the given parabola (1) is $y - 4 = 0$

(iii) Equation of the Tangent at the Vertex: Equation of the tangent at the vertex of the parabola (3) is $X = 0 \Rightarrow x + 1 = 0$

So, the equation to the tangent at the vertex of the given parabola (1) is $x + 1 = 0$

(iv) Equation of the Latus Rectum: Equation or rectum of the parabola (3) is

$X = 3 \Rightarrow x + 1 = 3 \Rightarrow x - 2 = 0$.

(vi) Coordinates of the Focus: The coordinates of the focus of the parabola (3) are (3, 0) which is equivalent to the simultaneous equations $x + 1 = 3$ and $y - 4 = 0$

$\Rightarrow x = 2, y = 4$

So, the coordinates of the focus of the given parabola (1) are (2,4)

(vii) Parametric Equations: The parametric equations of the parabola (3) are

$X = 3t^2$ and $Y = 6t$

$\Rightarrow x + 1 = 3t^2$ and $y - 4 = 6t$

or $x = -1 + 3t^2$ and $y = 4 + 6t$

Thus, the given parabola (1) has parametric equation:

$x = -1 + 3t^2, y = 4 + 6t$

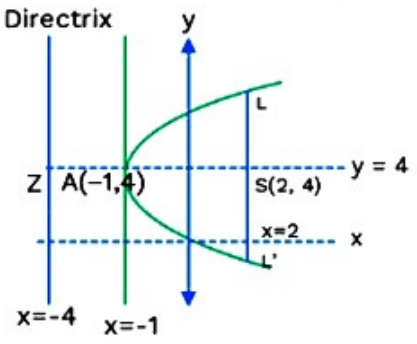
(viii) Length of the Latus Rectum : The length of the latus rectum of the parabola (3) is 12.

Since, the length will be independent of the choice of the origin. So, the length of the latus rectum of the given parabola is 12.

We can understand it more clearly from the following table :

Proposition	X, Y System	x, y System
Equation of Directrix	$X = -3$	$x + 1 = -3 \Rightarrow x + 4 = 0$
Equation of Axis	$Y = 0$	$y - 4 = 0 \Rightarrow y = 4$
Equation of Tangent at Vertex	$X = 0$	$x + 1 = 0$
Equation of Latus Rectum	$X = 3$	$x + 1 = 3 \Rightarrow x = 2$
Coordinates of Vertex	(0, 0)	$x + 1 = 0$ and $y - 4 = 0$ $\Rightarrow x = -1$ and $y = 4 \Rightarrow (-1, 4)$
Coordinates of Focus	(3, 0)	$x + 1 = 3$ and $y - 4 = 0 \Rightarrow x = 2$ and $y = 4 \Rightarrow (2, 4)$
Parametric Equations	$X = 3t^2$ $Y = 6t$	$x + 1 = 3t^2 \Rightarrow x = -1 + 3t^2$ $y - 4 = 6t \Rightarrow y = 4 + 6t$
Length of Latus Rectum	12	12

The graph of the given parabola is



(B) The given equation is

$4x^2 + 12x - 16y + 41 = 0$ ... (4)

or $x^2 + 3x - 4y + \frac{41}{4} = 0$

or $x^2 + 3x + \frac{9}{4} - 4y + 8 = 0$ or $\left(x + \frac{3}{2}\right)^2 = 4(y - 2)$ ... (5)

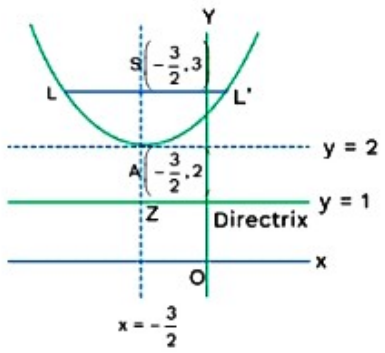
Shifting the origin to $\left(-\frac{3}{2}, 2\right)$ and using transformation equations

$x = X - \frac{3}{2}$ and $y = Y + 2$, we get from equation (5)

$X^2 = 4Y$ ... (6)

We are using the table to write the various propositions for the transformed equation (6) and for the original equation (4) by using retransformation equations:

PROPOSITION	X, Y SYSTEM	X, Y SYSTEM
Equation of Directrix	Y=-1	y-2=-1⇒y-1=0
Equation of Axis	X=0	x+3/2=0⇒2x+3=0
Equation of Tangent at Vertex	Y=0	y-2=0⇒y=2
Equation of Latus Rectum	y=1	y-2=1⇒y-3=0
Coordinates of Focus	(0,1)	x+3/2=0⇒x=-3/2 y-2 = 1 ⇒ y = 3 $\left(\frac{-3}{2}, 3\right)$
Parametric Equations	X = 2t Y = t^2	$x + \frac{3}{2} = 2t \Rightarrow x = -\frac{3}{2} + 2t$ $y - 2 = t^2 \Rightarrow y = 2 + t^2$



Q. **Find equation of parabola passing through (-4,-7), axis is || to x- axis and vertex at point (4,-3)**

Sol. If Parabola has to pass through (–4, –7), following case is possible.

$(y + 3)^2 = 4a(x - 4)$

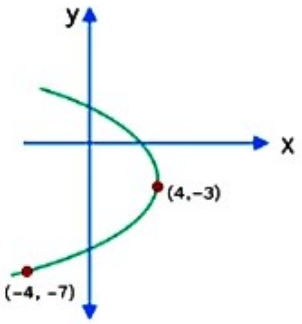
(–4, – 7) should satisfy this equation.

$16 = 4a(-8)$

$4a = -2$

Hence equation

$(y + 3)^2 = -2(x - 4)$



Q. **Determine the equation of the parabola having vertex (–3, 0) and directrix is x + 5 = 0**

Sol. A line passing through the vertex (–3, 0) and perpendicular to directrix x + 5 = 0 is x-axis. Thus, we can say that x-axis is the axis of the parabola by definition. Now, take the focus of the parabola as (a, 0). Since vertex, is the mid-point of Z(–5, 0) and focus S, therefore

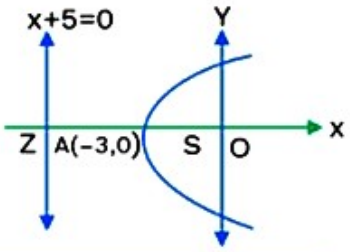
$$-3 = \frac{(a-5)}{2} \Rightarrow a = -1$$

$\therefore$ Focus = (–1, 0)

Thus the equation to the parabola is

$(x + 1)^2 + y^2 = (x + 5)^2$

$\Rightarrow y^2 = 8(x + 3)$



Q. **Find the equation of directrix and axis of the parabola $4y^2 - 6x - 4y = 5$.**

Sol. Here $4y^2 - 4y = 6x + 5$

$$4\left(y - \frac{1}{2}\right)^2 = 6(x+1) \Rightarrow \left(y - \frac{1}{2}\right)^2 = \frac{3}{2}(x+1)$$

Put $y - \frac{1}{2} = Y$ & $x + 1 = X$

The equation in standard form $Y^2 = \frac{3}{2}X$

$4a = \frac{3}{2} \Rightarrow a = \frac{3}{8}$ & length of latus rectum $= \frac{3}{2}$

Directrix, X + a = 0

$\Rightarrow x+1+\frac{3}{8}=0 \Rightarrow 8x+11=0$

Axis is $Y=0 \Rightarrow y-\frac{1}{2}=0 \Rightarrow 2y-1=0$

Parametric equation is $x+1=\frac{3}{8}t^2$ & $y-\frac{1}{2}=2\cdot\frac{3}{8}\cdot t$

i.e. $x=-1+\frac{3}{8}t^2, y=\frac{1}{2}+\frac{3}{4}t$.

Q. **Find the vertex, axis, focus, directrix, latus rectum of the parabola, also draw their rough sketches. $4y^2 + 12x - 20y + 67 = 0$**

Sol. The given equation is $4y^2+12x-20y+67=0 \Rightarrow y^2+3x-5y+\frac{67}{4}=0$

$\Rightarrow y^2-5y=-3x-\frac{67}{4} \Rightarrow y^2-5y+\left(\frac{5}{2}\right)^2=-3x-\frac{67}{4}+\left(\frac{5}{2}\right)^2$

$\Rightarrow \left(y-\frac{5}{2}\right)^2=-3x-\frac{42}{4} \Rightarrow \left(y-\frac{5}{2}\right)^2=-3\left(x+\frac{7}{2}\right)$ ... (i)

Let $X=x+\frac{7}{2}$ & $Y=y-\frac{5}{2}$ ... (ii)

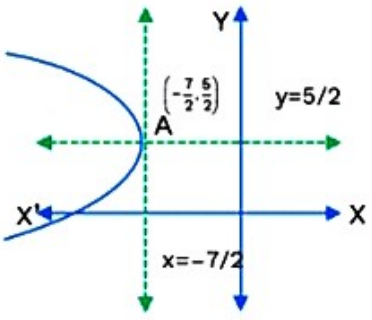
Using these relations, equation (i) reduces to $Y^2 = -3X$ .... (iii)
This is of the form $Y^2 = -4aX$ On comparing, we get $4a = 3 \Rightarrow a = 3/4$.
(X = 0, Y = 0) is the coordinates of the vertex,
i.e., the coordinates of the vertex are $\left(-\frac{7}{2},\frac{5}{2}\right)$

Y = 0 is the equation of the axis of the parabola,
i.e., the equation of the axis is $y=\frac{5}{2}$

[Putting Y = 0 in (ii)]
The coordinates of the focus are (X = –a, Y = 0)

$\Rightarrow x+\frac{7}{2}=-\frac{3}{4}$ & $y-\frac{5}{2}=0$

i.e., the coordinates of the focus are (–17/4, 5/2)



The equation of the directrix is represented by X = a i.e., $X=\frac{3}{4}$,

i.e., the equation of the directrix is $x=-\frac{11}{4}$

The length of the latus rectum of this parabola is 4a = 3.

Q. **Find the equation to the parabola whose focus is S(–1, 1) and directrix is 4x + 3y – 24 = 0 . Also find its axis, the vertex, the length and the equation of the latus rectum.**

Sol. Let P(x, y) be any point lying on the parabola. Since the distance of P from the focus is equal to its distance from the directrix,

i.e., PS = PQ or $PS^2 = PQ^2$

or $(x+1)^2 + (y-1)^2 = \left[\frac{(4x+3y-24)}{5}\right]^2$

i.e., $9x^2 + 16y^2 - 24xy + 242x + 94y - 526 = 0$

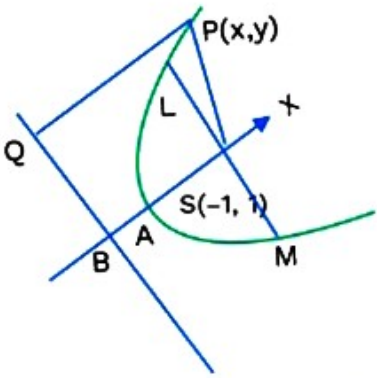
Above equation is the needed equation of the parabola.

The axis is a line through S(–1, 1) and $\perp$ to the directrix 4x + 3y – 24 = 0. Thus the equation of the axis is 3(x + 1) – 4(y – 1) = 0 or 3x – 4y + 7 = 0

The axis and the directrix intersect at B.

Solving them, we get B(3, 4)

The vertex A is the midpoint of S(–1, 1) and B(3, 4) Thus vertex A is $\left(1, \frac{5}{2}\right)$



Also $AS = \sqrt{\left[2^2 + \left(\frac{3}{2}\right)^2\right]} = \frac{5}{2}$

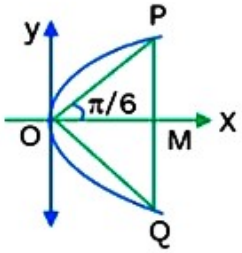
Hence length of the latus rectum 4AS = 10

Now, latus rectum is a straight line through the focus S and parallel to the directrix.

Hence, its equation is 4x + 3y + 1 = 0

Q. **An equilateral triangle has been inscribed in the parabola $y^2 = 4ax$, such that one vertex of this triangle is actually the vertex of the parabola. Then find the side length of this triangle.**

Sol. If ΔOPQ is an equilateral triangle then OP = OQ = PQ = k



$\Rightarrow \angle POM = \frac{\pi}{6}$

$\therefore$ $OM = k\cos\frac{\pi}{6}$ and $PM = k\sin\frac{\pi}{6}$

$\therefore$ $P\left(\frac{\sqrt{3}}{2}k, \frac{k}{2}\right)$ lie on the parabola $y^2 = 4ax$

$\Rightarrow \left(\frac{k}{2}\right)^2 = 4a \cdot \frac{\sqrt{3}}{2}k \Rightarrow k = 8\sqrt{3}a$

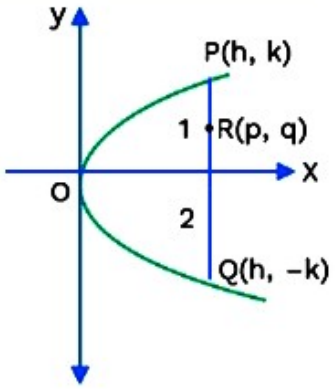
Q. **Find Locus of point of trisection of double ordinate of parabola $y^2 = 4ax$**

Sol. Let R(p, q) divides PQ in ratio 1 : 2

$p = h \qquad q = \frac{-k + 2k}{3}$

$q = \frac{k}{3}$

Now (h, k) lies on parabola
hence it satisfies its equation
$k^2 = 4ah$
$(3q)^2 = 4ap$
$9q^2 = 4ap$
hence locus is
$9y^2 = 4ax$



Q. **A parabola is parametrically represented by $x = t^2 + 1, y = 2t + 1$. Then find the coordinate of vertex and length of latus-rectum.**

Sol. Eliminate t from parametric equation, we get equation of parabola.

Hence, $x = \left(\frac{y-1}{2}\right)^2 + 1$ or $(y-1)^2 = 4(x-1)$

$\therefore$ vertex is (1, 1) and length of latus rectum = 4.

Q. **Find the parametric equation of the parabola $(x - 1)^2 = -12(y - 2)$**

Sol. $\because 4a = -12 \Rightarrow a = -3$

parametric equation is $y - 2 = -3t^2$

$x - 1 = -6t \qquad \Rightarrow x = 1 - 6t, y = 2 - 3t^2$

Q. **The parametric equation of the curve $(y-2)^2 = 12(x-4)$ are-**

(A) $6t, 3t^2$ **(B) $2 + 3t, 4 + t^2$**
(C) $4 + 3t^2, 2 + 6t$ **(D) None of these**

Sol. **(C)**

Here a = 3
$x - 4 = at^2 \Rightarrow x = 4 + 3.t^2 = 4 + 3t^2$
$y - 2 = 2at \Rightarrow y = 2 + 2.3t = 2 + 6t$

Q. **Find the equation of parabola, whose axis is parallel to y -axis and which passes through points (0, 2), (-1, 0) and (1, 6)**

Sol. General equation of such parabola is $y = Ax^2 + Bx + C$ ... (i)

Since it passes through $(0,2)(-1,0)$ and $(1,6)$, then we have

$C = 2$ ... (ii)

$A - B + C = 0$ .... (iii)

$A + B + C = 6$ .... (iv)

from (ii), (iii) and (iv),

$A = 1, B = 3, C = 2$

$\therefore$ equation of parabola is $y = x^2 + 3x + 2$ Ans.

Q. **Find the value of λ if the equation $\sqrt{(x-1)^2 + (y-2)^2} = \sqrt{\lambda}(x + y + 3)$ represents a parabola. Find its focus, directrix, axis and equation of vertex.**

Sol. $(x-1)^2 + (y-2)^2 = \lambda(x + y + 3)^2$ are given equation.

$$(x-1)^2 + (y-2)^2 = 2\lambda\left(\frac{x + y + 3}{\sqrt{2}}\right)^2$$

$\Rightarrow$ above equation represents parabola if $2\lambda = 1 \Rightarrow \lambda = \frac{1}{2}$

In this case focus is $(1,2)$ and directrix is $x + y + 3 = 0$

Q. **AP is perpendicular to PB, where A is the vertex of parabola $y^2 = 4x$ and P is on the parabola. If B lie on the axis of parabola, then find the locus of centroid of ΔPAB**

Sol. Slope of $AP = \frac{2}{t}$

$\therefore$ Slope of $BP = -\frac{t}{2}$

$\therefore$ equation of line PB is $y - 2t = -\frac{t}{2}\left(x - t^2\right)$

$\therefore$ point B is $\left(t^2 + 4, 0\right)$

Let centroid of ΔPAB is (h, k)

$$h = \frac{t^2 + t^2 + 4}{3} \text{ and } k = \frac{2t}{3}$$

Now eliminate t from above two equation, we get

$$3h - 4 = 2\left(\frac{3k}{2}\right)^2$$

locus of centroid is $3x - 4 = \frac{9y^2}{2}$.

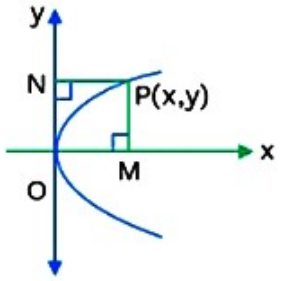
Equation of parabola with respect to two given perpendicular lines :

Let P(x, y) be any point on the parabola. Then we can rewrite the equation of parabola $y^2 = 4ax$ as $(PM)^2 = 4a(PN)$

i.e., (The distance of P from its axis)^2

= (length of latus-rectum) × (The distance of P from the tangent at its vertex)

where P is any point on the parabola.



Q. **Find the equation of the parabola whose latus rectum is 4 units, axis is the line 3x + 4y – 4 = 0 and the tangent at the vertex is the line** 4x – 3y + 7 = 0.

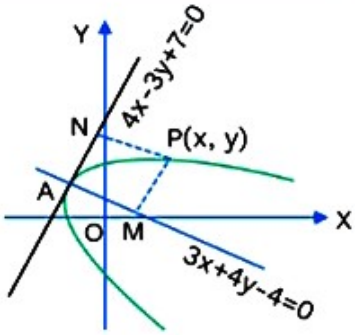
Sol. Let P(x, y) be any point lying on the parabola and let PM and PN are perpendiculars from P on the axis and tangent at the vertex respectively then

$(PM)^2 =$ (latus rectum) (PN)

$$\Rightarrow \left(\frac{3x + 4y - 4}{\sqrt{3^2 + 4^2}}\right)^2 = 4\left(\frac{4x - 3y + 7}{\sqrt{4^2 + (-3)^2}}\right)$$

$$\Rightarrow (3x + 4y - 4)^2 = 20(4x - 3y + 7)$$

which is the needed equation of parabola.



Q. **Variable circle is described to pass through point (1, 0) and tangent to the curve y = tan(tan^{-1}x). The locus of the centre of the circle is a parabola whose:**

(A) Length of the latus rectum is $2\sqrt{2}$

(B) Axis of symmetry is represented by the equation x + y = 1

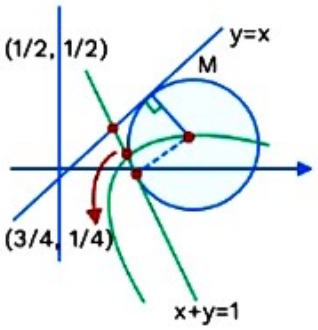
(C) Vertex has the co-ordinates (3/4, 1/4)

(D) None of these

Sol. **(BC)**

Distance of centre from (1,0) is equal to distance of centre from line y=x. Hence, y = x is directrix and (1,0) is focus.
Axis will be y+x=1 and vertex has the co-ordinates (3/4, 1/4).

$$4a = 2\left(\frac{|0-1|}{\sqrt{2}}\right) = \frac{2}{\sqrt{2}} = \sqrt{2}$$



Q. **If a variable circle touches a fixed circle and a fixed line then prove that the locus of the centre of the variable circle is a parabola whose directrix is parallel to a given line at a distance equal to the radius of the given circle.**

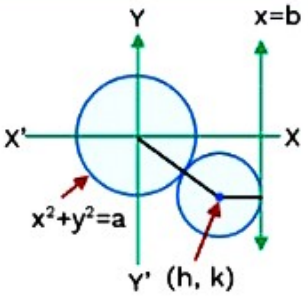
Sol. Let given circle be $x^2 + y^2 = a^2$ and given line be x = b.
From the diagram radius of variable circle is b – h.
If it touches $x^2 + y^2 = a^2$,

then $a + (b - h) = \sqrt{h^2 + k^2}$

$\Rightarrow (a + b)^2 - 2(a + b)h + h^2 = h^2 + k^2$

$\Rightarrow y^2 = (a + b)^2 - 2(a + b)x$

which is equation of parabola.



Q. **Prove that the area of the triangle whose vertices are (x_i, y_i), i = 1, 2, 3 and inscribed in the parabola $y^2 = 4ax$ is** $\frac{1}{8a}\left|(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)\right|$

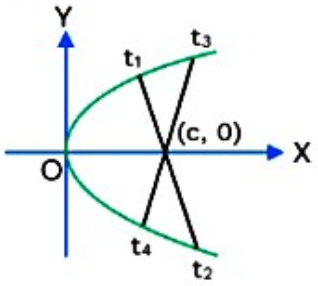
Sol. Let x_1, x_2, and x_3 be the abscissae of the points on the parabola whose ordinates are y_1, y_2, and y_3, respectively. Then $y_1^2 = 4ax_1$, $y_2^2 = 4ax_2$, and $y_3^2 = 4ax_3$. Therefore, the area of the triangle having the vertices are $(x_1, y_1), (x_2, y_2)$, and (x_3, y_3) is

$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} y_1^2/4a & y_1 & 1 \\ y_2^2/4a & y_2 & 1 \\ y_3^2/4a & y_3 & 1 \end{vmatrix} = \frac{1}{8a}\begin{vmatrix} y_1^2 & y_1 & 1 \\ y_2^2 & y_2 & 1 \\ y_3^2 & y_3 & 1 \end{vmatrix} = \frac{1}{8a}\left|(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)\right|$$

Chord

A Chord is a line joining any two points on the parabola.

Let the points $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$, lie on the parabola then equation of chord is



$$(y - 2at_1) = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2}\left(x - at_1^2\right)$$

$$\Rightarrow y - 2at_1 = \frac{2}{t_1 + t_2}\left(x - at_1^2\right)$$

$$\Rightarrow (t_1 + t_2)y = 2x + 2at_1t_2$$

If this chord meet the x-axis at point $(c, 0)$ then from above equation

$-c = at_1t_2 = 0$ i.e. $t_1 t_2 = -c/a$

Point to Remember!!!

(i) If the chord joining t_1, t_2 & t_3, t_4 pass through a point $(c, 0)$ on the axis, then $t_1t_2 = t_3t_4 = -c/a$

.

(ii) If PQ is a focal chord then $t_1t_2 = -1$ or $t_2 = -\frac{1}{t_1}$. which is required relation.

Hence if a focal chord has one of the extremity as $(at^2, 2at)$ then the other extremity will be $\left(\frac{a}{t^2}, -\frac{2a}{t}\right)$.

Q. **A quadrilateral has been inscribed in a parabola $y^2 = 4ax$ with three of its sides passing through the fixed points on the axis. Prove that the fourth side also passes through fixed point on the axis of the parabola.**

Sol. Equation of chord AB is

$2x - (t_1 + t_2)y + 2at_1t_2 = 0$

It passes through the point (c, 0),

then given $t_1t_2 = -\frac{c}{a}$

Similarly for chord BC, $t_2t_3 = -\frac{d}{a}$

And for chord CD, $t_3t_4 = -\frac{e}{a}$

Multiplying (i) and (iii), we get

$$t_1t_2t_3t_4 = +\frac{ec}{a^2}$$

$$\Rightarrow \quad t_1t_4\left(-\frac{d}{a}\right) = \frac{ec}{a^2}$$

$$\Rightarrow \quad t_1t_4 = -\frac{ec}{ad}$$

Hence, chord AD passes through the fixed point.

Q. **Show that the focal chord of parabola $y^2 = 4ax$ makes an angle α with the x -axis, then its length is equal to** $4a\text{cosec}^2\alpha$

Sol. Let $P\left(at_1^2, 2at_1\right)$ and $Q\left(at_2^2, 2at_2\right)$ be the end points of a focal chord PQ which makes an angle α with the axis of the parabola. Then

$$PQ = a\left(t_2 - t_1\right)^2$$

$$= a\left(t_2 + t_1\right)^2 - 4t_1t_2\Big)$$

$$= a\left(\left(t_2 + t_1\right)^2 + 4\right) \quad \left(\because t_1t_2 = -1\right). \qquad (\because t_1t_2 = -1) \qquad \ldots (1)$$

$$\tan\alpha = \text{ slope of } PQ = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2} = \frac{2}{t_2 + t_1}$$

$$t_2 + t_1 = 2\cot\alpha \qquad \ldots (2)$$

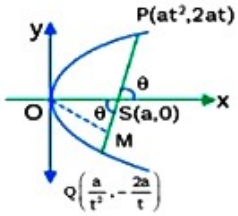
Substituting the value of $t_2 + t_1$ from (2) in (1) then

$$PQ = a\left(4\cot^2\alpha + 4\right) = 4a\text{cosec}^2\alpha$$

Q. **If the length of a focal chord of the parabola $y^2 = 4ax$ at a distance b from the vertex is c, then show that $b^2c = 4a^3$**

Sol. OM = b

focal length $PQ = 4a\text{cosec}^2\theta$ ... (i)



From right angled triangle OMS,

$$\sin\theta = \frac{OM}{OS} = \frac{b}{a}$$

$$\therefore \quad \text{cosec}\theta = \frac{a}{b} \qquad \ldots(ii)$$

from (i) and (ii)

$$PQ = c = 4a \cdot \left(\frac{a}{b}\right)^2$$

$$\Rightarrow b^2c = 4a^3$$

Q. **If the point $(at^2, 2at)$ be the extremity of a focal chord of parabola $y^2 = 4ax$ then show that the length of the focal chord is $a\left(t + \frac{1}{t}\right)^2$**

Sol. Since one extremity of focal chord is $P(at^2, 2at)$ then the other extremity will be

$$Q\left(\frac{a}{t^2}, -\frac{2a}{t}\right) \qquad \text{(Replacing t by } -1/t)$$

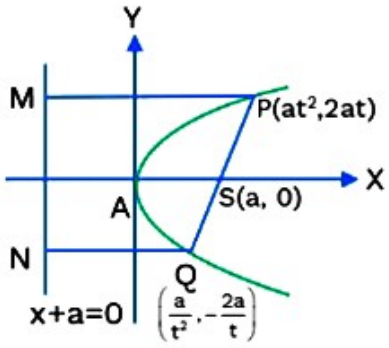
$$= SP + SQ \qquad (\because SP = PM \text{ and } SQ = QN)$$

$$= PM + QN$$

$$= at^2 + a + \frac{a}{t^2} + a$$

$$= \left(t^2 + \frac{1}{t^2} + 2\right)a$$

$$= a\left(t + \frac{1}{t}\right)^2$$



Note :

(i) The length of focus chord having parameters t_1 and t_2 for its and points is $a(t_2 - t_1)^2$.

(ii) $\because \left|t + \frac{1}{t}\right| \geq 2$ for all $t \neq 0$ $(\because AM \geq GM)$

$$a\left(t + \frac{1}{t}\right)^2 \geq 4a$$

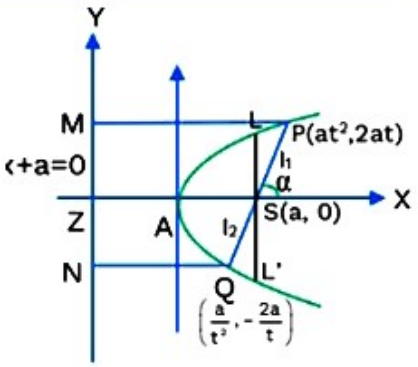
$\Rightarrow$ Length of a focal chord is always greater than or equal to length of the latus rectum i.e., the latus rectum of a parabola can also be called the Shortest focal chord of a parabola.

Q. **Show that the semi latus rectum of the parabola $y^2 = 4ax$ and the segements of any focal chord of the parabola forms a Harmonic Progression.**

Sol. Let parabola be $y^2 = 4ax$

If PQ be the focal chord then

$P = (at^2, 2at)$ and $Q = \left(\frac{a}{t^2}, \frac{-2a}{t}\right)$



If segment of focal chord are l_1 and l_2

then $l_1 = SP = PM = a + at^2 = a(1 + t^2)$

and $l_2 = SQ = QN = a + \frac{a}{t^2} = \frac{a(1+t^2)}{t^2}$

$\therefore$ Harmonic mean of l_1 and l_2

$$= \frac{2l_1l_2}{l_1 + l_2} = \frac{2}{\frac{1}{l_2} + \frac{1}{l_1}} = \frac{2}{\frac{t^2}{a(1+t^2)} + \frac{1}{a(1+t^2)}} = \frac{2}{1/a} = 2a = \text{Semi latus rectum.}$$

Note:

If the length of segments of a focal chord of a parabola are l_1 and l_2, then length of the latus rectum of this parabola is $\frac{4l_1l_2}{l_1 + l_2}$.

Q. **All chords of the parabola $y^2 = 4ax$ subtending a right angle at the vertex passes through a fixed point (4a, 0).**

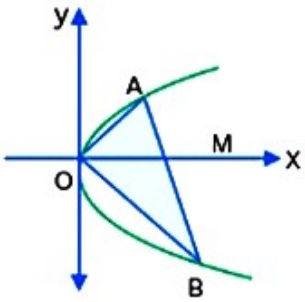
Sol. Let $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$ be the extremities of a chord of a parabola

$y^2 = 4ax$.

The vertex of the parabola $y^2 = 4ax$ is O(0,0)

Slope of the line OA is

$$m_1 = \frac{2at_1}{at_1^2} = \frac{2}{t_1}$$



Slope of the line OB is $m_2 = \frac{2at_2}{at_2^2} = \frac{2}{t_2}$

If AB subtends right angle at O, then

$m_1m_2 = -1$

$\Rightarrow \frac{2}{t_1} \cdot \frac{2}{t_2} = -1$ or $t_1t_2 = -4$.

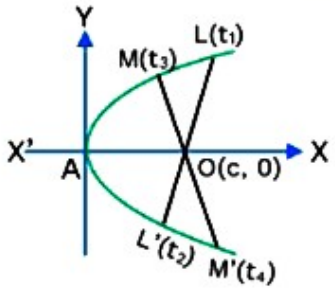
Now the equation of the chord AB is
$(t_1 + t_2)y = 2x + 2at_1t_2$
$\Rightarrow (t_1 + t_2)y = 2x + 2a(-4)$ [$\because t_1t_2 = -4$]

or $y = \frac{2}{t_1 + t_2}(x - 4a)$

Clearly, the above line passes through the point (4a, 0) for all values of t_1 and t_2. Thus, the chords which subtend right angle at the vertex of parabola pass through the fixed point (4a, 0), which lies on the axis of the curve.

Q. **LOL' and MOM' are two chords of parabola $y^2 = 4ax$ having vertex A and passing through a point O on its axis. Show that the radical axis of the circles described on LL' and MM' as diameters passes through the vertex of the parabola.**

Sol. Equation of LOL : $2x - (t_1 + t_2)y + 2at_1t_2 = 0$
Since chord LOL' passes through (c, 0)



$\therefore\ t_1t_2 = -\frac{c}{a}$

Similarly, for chord MOM' $t_3t_4 = -\frac{c}{a}$

Now circle with LL ' as diameter, the equation is

$(x - at_1^2)(x - at_2^2) + (y - 2at_1)(y - 2at_2) = 0$

or $x^2 + y^2 - a(t_1^2 + t_2^2)x - 2a(t_1 + t_2)y + c^2 - 4ac = 0$ ...(i)

(as $a^2t_1t_2^2 = c^2$ and $4a^2t_1t_2 = -4ac$)

Similarly, circle with MM' as diameter, the equation is

$x^2 + y^2 - a(t_3^2 + t_4^2)x - 2a(t_3 + t_4)y + c^2 - 4ac = 0$ ...(ii)

Radical axis of Eqs. (i) and (ii) is (i) - (ii) or

$a(t_1^2 + t_2^2 - t_3^2 - t_4^2)x - 2a(t_1 + t_2 - t_3 - t_4)y = 0$,

which passes through the origin (vertex of the parabola).

Q. **Show that on the axis of any parabola there exists a point K such that if a chord PQ of the parabola is drawn through it, then $\frac{1}{PK^2} + \frac{1}{QK^2}$ is the same for all positions of the chord.**

Sol. Let the point K on the axis of the parabola $y^2 = 4ax$ has the coordinates (h, 0) Equation of the line through K at an angle θ with the axis, is

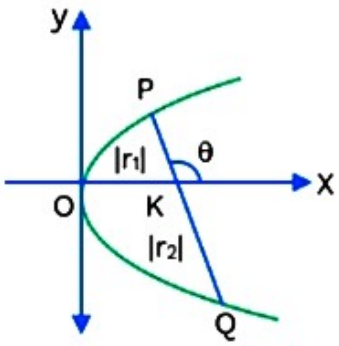
$$\frac{x-h}{\cos\theta} = \frac{y-0}{\sin\theta} = r \qquad \text{...(i)}$$

Any point on this line can be selected as $(h + r\cos\theta, r\sin\theta)$. This point if lies on the parabola $y^2 = 4ax$, then

$$(r\sin\theta)^2 = 4a(h + r\cos\theta) \text{ or } (\sin^2\theta)r^2 - 4a\cos\theta r - 4ah = 0 \qquad \text{...(ii)}$$

The roots r_1 and r_2 of the equation (2) represent the algebraic distances PK and OK. We have

$$r_1 + r_2 = \frac{4a\cos\theta}{\sin^2\theta} \text{ and } r_1 r_2 = -\frac{4ah}{\sin^2\theta}$$



$$\text{Now, } \frac{1}{PK^2} + \frac{1}{QK^2} = \frac{1}{r_1^2} + \frac{1}{r_2^2} = \frac{(r_1 + r_2)^2 - 2r_1 r_2}{r_1^2 r_2^2}$$

$$= \frac{\left(\frac{4a\cos\theta}{\sin^2\theta}\right)^2 + 2\cdot\frac{4ah}{\sin^2\theta}}{\left(\frac{-4ah}{\sin^2\theta}\right)^2}$$

$$= \frac{16a^2\cos^2\theta + 8ah\sin^2\theta}{16a^2h^2} = \frac{2a + (h-2a)\sin^2\theta}{2ah^2}$$

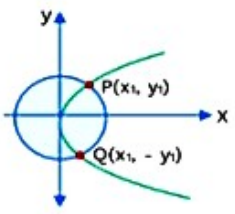
If we select h = 2a, then above expression becomes independent of θ and we get $\frac{1}{PK^2} + \frac{1}{QK^2} = \frac{2a}{2a(2a)^2} = \frac{1}{4a^2}$

$\frac{1}{PK^2} + \frac{1}{QK^2}$ is constant for any direction of the chord PQ through k

Q. **A circle is described whose centre is the vertex and whose diameter is three quarters of the latus rectum of a parabola. Show that the common chord of the circle and parabola bisects the distance between the vertex and the focus.**

Sol. Let parabola $y^2 = 4ax$
equation of circle will be
$x^2 + y^2 = 9a^2/4$
Solving P and Q will be

$$\left(\frac{a}{2}, \sqrt{2}a\right) \text{ and } \left(\frac{a}{2}, -\sqrt{2}a\right)$$



equation of common chord $x = a/2$
which bisects (0,0) and (a, 0)

Q. **A circle and a parabola $y^2 = 4ax$ intersect in four points; prove that the ordinates of the four points have their algebraic sum equal to zero. Also show that the line joining one pair of these four points is equally inclined to the axis with the line joining the other pair.**

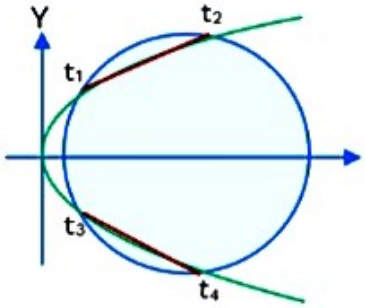
Sol. $x^2 + y^2 + 2gx + 2fy + c = 0$
Solving it with
$a^2t^4 + 4a^2t^2 + 2gat^2 + 4aft + c = 0$
$a^2t^4 + 2a(2a + g)t^2 + 4aft + c = 0$
Hence, $t_1 + t_2 + t_3 + t_4 = 0$ ... (i)
or $2a(t_1 + t_2 + t_3 + t_4) = 0$
Hence, proved
Slope of line joining

$$t_1, t_2 = \frac{2}{t_1 + t_2} = -\frac{2}{t_3 + t_4}$$

$= m_1$ [Using Eq. (i)]

Slope of line joining $t_3, t_4 = \frac{2}{t_3 + t_4} = m_2$

Hence, $m_1 + m_2 = 0$



Position of a point with respect to a parabola $y^2 = 4ax$

Let $P(x_1, y_1)$ be a point. From P draw $PM \perp AX$ (on the axis of parabola) meeting the parabola $y^2 = 4ax$ at $Q(x_1, y_2)$ where $Q(x_1, y_2)$ lie on the parabola therefore

$y_2^2 = 4ax_1$ ... (1)

Now, P will be inside, on or outside the parabola.

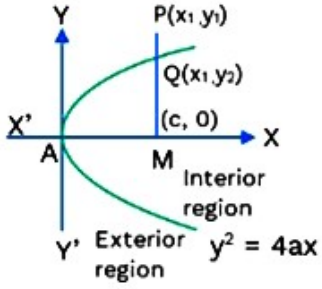
$y^2 = 4ax$ according as

$PM >, =,$ or $< QM$

$\Rightarrow (PM)^2 >, =,$ or $< (QM)^2$

$\Rightarrow y_1^2 >, =,$ or $< y_2^2$

$\Rightarrow y_1^2 >, =,$ or $< 4ax_1$ (from (1))



Hence $y_1^2 - 4ax_1 >, =,$ or < 0

Hence, let us consider the equation of parabola $S(x, y) = y^2 - 4ax$

(i) If $S(x_1, y_1) > 0$ then $P(x_1, y_1)$ lie outside the parabola.

(ii) If $S(x_1, y_1) < 0$ then $P(x_1, y_1)$ lie inside the parabola.

(iii) If $S(x_1, y_1) = 0$ then $P(x_1, y_1)$ lie on the parabola.

Know the facts



This result also holds true for circle, parabola and ellipse.

Q. **Show that the point (2, 3) lies outside the parabola $y^2 = 2x$.**

Sol. Let $S(x, y) \equiv y^2 - 2x$
$\therefore S(2, 3) = 9 - 2.2 = 5 =$ positive
(2, 3) lie outside the parabola

Q. **Find the position of the point (–2, 2) with respect to the parabola $y^2 - 4y + 9x + 11 = 0$**

Sol. Let $S(x, y) = y^2 - 4y + 9x + 11$
$\therefore S(-2, 2) = -11 =$ negative

$\Rightarrow$ (–2, 2) lie inside the parabola

INTERACTION BETWEEN THE LINE AND PARABOLA

Consider the equation of the given line be $y = mx + c$ ...(i)

and equation of the parabola be $y^2 = 4ax$ ...(ii)

then line may cut, touch or does not meet parabola.

The points of intersection of the line (1) and the parabola (2) will be obtained by solving the two equations simultaneously. By solving equation (i) and (ii), we get

$my^2 - 4ay + 4ac = 0$

There are two roots of above equation.

The nature of the roots will be decided by the

discriminant $D = 16a(a - cm)$

(i) Now, if $D > 0$ i.e., $c < \frac{a}{m}$, then line intersect the parabola at two distinct points.

(ii) If $D = 0$ i.e., $c = \frac{a}{m}$, then line touches the parabola. (It is condition of tangency)

(iii) If $D < 0$ i.e., $c > \frac{a}{m}$, then line neither touch nor intersect the parabola.

EQUATION OF TANGENT

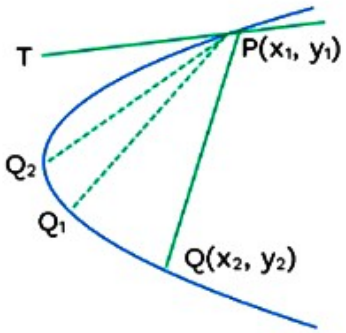
1. Point Form :

Equation of parabola is $y^2 = 4ax$ ... (1)

Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be any two points on parabola (1), then

$y_1^2 = 4ax_1$ ... (2)

and $y_2^2 = 4ax_2$ ... (3)



Subtracting (2) from (3) then

$y_2^2 - y_1^2 = 4a(x_2 - x_1)$

or $\frac{y_2 - y_1}{x_2 - x_1} = \frac{4a}{y_2 + y_1}$ ... (4)

Equation of PQ is

$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$ ... (5)

From (4) and (5), then

$y - y_1 = \frac{4a}{y_2 + y_1}(x - x_1)$ ... (6)

Now to find tangent at P, Q → P, i.e., $x_2 \to x_1$ and $y_2 \to y_1$ then equation (6) becomes

$y - y_1 = \frac{4a}{2y_1}(x - x_1)$

or $yy_1 - y_1^2 = 2ax - 2ax_1$

or $yy_1 = 2ax + y_1^2 - 2ax_1$

or $yy_1 = 2ax + 4ax_1 - 2ax_1$ [from (2)]

or $yy_1 = 2ax + 2ax_1$

which is the equation of tangent at (x_1, y_1)

The equation of tangent at (x_1, y_1) can also be obtained by replacing

x^2 by xx_1, y^2 by yy_1, x by $\frac{x + x_1}{2}$, y by $\frac{y + y_1}{2}$ and xy by $\frac{xy_1 + yx_1}{2}$ in the equation of curve. The constant (if any) in the equation remains as it was. This method (standard substitution) is applicable to all conic when point lies on the conic.

Equation of tangent of standard parabola :

Equation of Parabolas	Tangent at (x_1, y_1)
$y^2 = 4ax$	$yy_1 = 2a(x + x_1)$
$y^2 = -4ax$	$yy_1 = -2a(x + x_1)$
$x^2 = 4ay$	$xx_1 = 2a(y + y_1)$
$x^2 = -4ay$	$xx_1 = -2a(y + y_1)$

2. Slope Form :

The equation of tangent to the parabola $y^2 = 4ax$ at (x_1, y_1) is

$$yy_1 = 2a(x + x_1) \quad \ldots (1)$$

Let m is the slope of the tangent. Then

$m = \frac{2a}{y_1}$ or $y_1 = \frac{2a}{m}$ Since (x_1, y_1) lies on $y^2 = 4ax$ therefore

$y_1^2 = 4ax_1$ or $\frac{4a^2}{m^2} = 4ax_1 \Rightarrow x_1 = \frac{a}{m^2}$

Replacing the values of x_1 and y_1 in (1), we get

$$y = mx + \frac{a}{m} \quad \ldots (2)$$

Thus, tangent to the parabola $y^2 = 4ax$ is represented by $y = mx + \frac{a}{m}$ for all values of m, where m denotes the slope of the tangent and the co-ordinates of the point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$.

Thus $y = mx + c$ is the tangent of $y^2 = 4ax$ for all values of m if only if $c = \frac{a}{m}$ and

$(y - k) = m(x - h) + \frac{a}{m}$ is tangent to the parabola

$(y - k)^2 = 4a(x - h)$

The equation of tangent, condition of tangency and point of contact in terms of slope (m) of standard parabolas are shown below in the table.

Equation of parabola	Point of contact in terms of slope (m)	Equation of tangnet in terms of slope (m)	Condition of tangency for line y = mx + c
$y^2 = 4ax$	$\left(\frac{a}{m^2}, \frac{2a}{m}\right)$	$y = mx + \frac{a}{m}$	$c = \frac{a}{m}$
$y^2 = -4ax$	$\left(-\frac{a}{m^2}, -\frac{2a}{m}\right)$	$y = mx - \frac{a}{m}$	$c = -\frac{a}{m}$
$x^2 = 4ay$	$(2am, am^2)$	$y = mx - am^2$	$c = -am^2$
$x^2 = -4ay$	$(-2am, -am^2)$	$y = mx + am^2$	$c = am^2$

Alternative method:

Let the parabola be $y^2 = 4ax$ ... (1)

and the given line by $y = mx + c$ ... (2)

Putting the value of y from (2) in (1), we get

$m^2x^2 + 2x(mc - 2a) + c^2 = 0$ ... (3)

Since line $y = mx + c$ is a tangent to parabola $y^2 = 4ax$, there will not be distinct solution. So, above quadratic equation (3) will have equal roots. The condition for this is $4(mc-2a)^2 - 4m^2c^2 = 0$ (Discriminant of the quadratic equation = 0)

or $-4mca + 4a^2 = 0$ or $c = \frac{a}{m}$, which is the required condition of tangency.

Note :

If P lies outside the parabola, then Two tangents can be drawn from a point $P(\alpha, \beta)$ to a parabola.

3. Parametric Form :

Any point on the parabola $y^2 = 4ax$ can be represented parametrically by $(at^2, 2at)$ or 't'
We realize that the equation of tangent of the parabola $y^2 = 4ax$ at (x_1, y_1) is characterised by $yy_1 = 2a(x + x_1)$ ... (1)
Substituting x_1 by at^2 and y_1 by 2 at, then (1) becomes $y(2at) = 2a(x + at^2)$
$ty = x + at^2$

Point of intersection of tangents at any two points lying on the parabola :

Let the parabola be $y^2 = 4ax$ and two points on the parabola are

$P = (at_1^2, 2at_1)$ and $Q = (at_2^2, 2at_2)$

Equation of tangents at

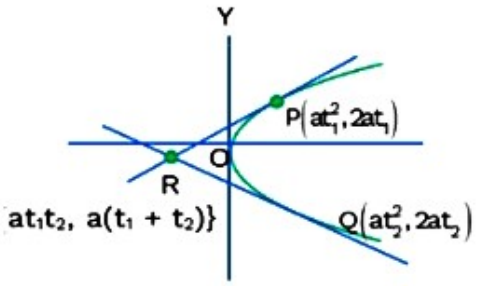
$P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$

are $t_1y = x + at_1^2$ ... (1)

and $t_2y = x + at_2^2$ ... (2)

Solving above equations, we get the values:
$x = at_1t_2$, $y = a(t_1 + t_2)$
Thus, the point of intersection of tangents have the co-ordinates

$P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ are $R(at_1t_2, a(t_1 + t_2))$



Point to Remember!!!

(i) The y - coordinate of the point of intersection of tangents at P and Q on the parabola represent the Arithmetic mean of the y-coordinates of P and Q

$\left(\text{i.e.}\ \frac{2at_1 + 2at_2}{2} = a(t_1 + t_2)\right)$

(ii) The x co-ordinate of the point of intersection of tangents at P and Q on the parabola represent the Geometric mean of the x-coordinates of P and Q

$\left(\text{i.e.}\ \sqrt{at_1^2 \times at_2^2} = at_1t_2\right)$

Q. **Show that the locus of the points of intersection of the tangents to a parabola at the extremity of focal chord are perpendicular and always meet at the directrix of the parabola.**

Sol. Let the points $P\left(at_1^2, 2at_1\right)$ and $Q\left(at_2^2, 2at_2\right)$ on the parabola $y^2 = 4ax$ and tangents at P and Q are

$t_1y = x + at_1^2$ ... (1)

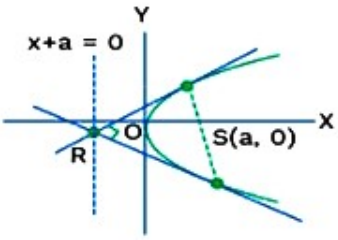
and $t_2y = x + at_2^2$ ... (2)

$\therefore$ point of intersection of these tangents is $\left(at_1t_2, a\left(t_1 + t_2\right)\right)$ Let this point is (h, k) then $h = at_1t_2$ ... (3)

and $k = a\left(t_1 + t_2\right)$ ... (4)

Slope of tangents (1) and (2) are $\frac{1}{t_1}$ and $\frac{1}{t_2}$ respectively.

$\therefore$ Product of slope $= \frac{1}{t_1} \times \frac{1}{t_2} = -1$



(Since PQ is focal chord)

$\Rightarrow$ h = – a from equation (3)

$\therefore$ Locus is x + a = 0 i.e. Directrix of parabola

Directrix is also called the Director Circle of the parabola.

Q. **If the line $2x - 3y = k$ touches the parabola $y^2 = 6x$, then find the value of k.**

Sol. Given $x = \frac{3y + k}{2}$ ... (1)

and $y^2 = 6x$ ... (2)

Solve (1) and (2), we get

$\Rightarrow y^2 = 6\left(\frac{3y + k}{2}\right) \Rightarrow y^2 = 3(3y + k)$

$\Rightarrow y^2 - 9y - 3k = 0$ ... (3)

If line (1) touches parabola (2) then roots of quadratic equation (3) is equal i.e. D = 0

$\therefore (-9)^2 = 4 \times 1 \times (-3k) \Rightarrow k = -\frac{27}{4}$

Alternative method :

Line $y = \frac{2}{3}x - \frac{k}{3}$ touch is $y^2 = 6x$ then use the condition of tangency

$c = \frac{a}{m} \Rightarrow -\frac{k}{3} = \frac{9}{4} \Rightarrow k = -\frac{27}{4}$

Q. **Find the equation to the tangents to the parabola $y^2 = 9x$ which goes through the point (4, 10).**

Sol. Equation of tangent to parabola $y^2 = 9x$ is

$y = mx + \frac{9}{4m}$

Since it passes through (4, 10)

$10 = 4m + \frac{9}{4m}$

$\Rightarrow 16m^2 - 40m + 9 = 0$

$m = \frac{1}{4}, \frac{9}{4}$

equation of tangent's are $y = \frac{x}{4} + 9$ & $y = \frac{9}{4}x + 1$

Q. **If the tangents at P and Q on a parabola (whose focus is S) meet in the point T, then prove that SP, ST and SQ are in geometric progression.**

Sol. Let $P\left(at_1^2, 2at_1\right)$ and $Q\left(at_2^2, 2at_2\right)$ be any two points on the parabola $y^2 = 4ax$, then point of intersection of tangents at P and Q will be

$T = \left[at_1t_2, a\left(t_1 + t_2\right)\right]$

Now $SP = a\left(t_J^2 + 1\right)$

$SQ = a\left(t_2^2 + 1\right)$

$ST = a\sqrt{\left(t_1^2 + 1\right)\left(t_2^2 + 1\right)}$ (use distance between two points formula)

$\therefore ST^2 = SP.\ SQ$

$\therefore$ SP, ST and SQ are in G.P.

Q. **If two tangents are drawn from the point (h, k) to the parabola $y^2 = 4x$ in such a way that slope of one tangent is double of the slope of other, then prove that $9h = 2k^2$**

Sol. Tangent to parabola $y^2 = 4x$ is

$y = mx + \frac{1}{m}$ and it passes through (h, k), so $k = mh + \frac{1}{m}$

i.e. $hm^2 - km + 1 = 0$

Its roots are m_1 and $2m_1$

$m_1 + 2m_1 = \frac{k}{h} \Rightarrow 3m_1 = \frac{k}{h}$ ...(i)

$m_1 \cdot 2m_1 = \frac{1}{h} \Rightarrow 2m_1^2 = \frac{1}{h}$ ...(ii)

from (i) and (ii) eliminate m, we get

$9h = 2k^2$

Q. **A tangent to the parabola $y^2 = 8x$ makes an angle of 45° with the straight line $y = 3x + 5$. Then find the equation of tangent and point of contact also.**

Sol. Here a = 2, equation of tangent at $(2t^2, 4t)$ is $yt = x + 2t^2$

slope of tangent $m = \frac{1}{t}$

$$\tan 45^\circ = \pm\frac{\frac{1}{t} - 3}{1 + \frac{1}{t} \cdot 3} \Rightarrow t = -\frac{1}{2} \text{ or } 2$$

when $t = -\frac{1}{2}$, then equation of tangent is $\left(-\frac{1}{2}\right) y = x + 2 \cdot \frac{1}{4}$

$\Rightarrow 2x + y + 1 = 0$ and point

of contact is $\left(\frac{1}{2}, -2\right)$. When t = 2, the tangent is $2y = x + 8$ and point of contact is $(2.\ (2)^2\ ,\ 4.\ 2)$ i.e. (8. 8).

Q. **Find the common tangent of the parabola $y^2 = 8ax$ and the circle $x^2 + y^2 = 2a^2$**

Sol. Any tangent to parabola is $y = mx + \frac{2a}{m}$

Solving with the circle $x^2 + \left(mx + \frac{2a}{m}\right)^2 = 2a^2$

$x^2\left(1+m^2\right) + 4ax + \left(\frac{4a^2}{m^2} - 2a^2\right) = 0$

The condition of tangency is $B^2 - 4AC = 0$ gives
$m^4 + m^2 - 2 = 0 \Rightarrow m = \pm 1$
Hence equation of tangent is $y = x + 2a$ and $y = -x - 2a$

Q. **Find equation of line touching both the parabola $y^2 = 4x$ and $x^2 = 32y$.**

Sol. Equation of tangent in terms of slope of parabola $y^2 = 4x$ is
$y = mx + \frac{1}{m}$

$\because$ Eq. (i) is also tangent of $x^2 = -32y$ then $x^2 = -32\left(mx + \frac{1}{m}\right)$

$\Rightarrow x^2 + 32mx + \frac{32}{m} = 0$

Above equation must have equal roots,
Hence its discriminant must be zero $\Rightarrow (32m)^2 = 4.1 \cdot \frac{32}{m}$

$\Rightarrow m^3 = \frac{1}{8}$ or $m = \frac{1}{2}$ From Eq. (i), $y = \frac{x}{2} + 2$

$\Rightarrow x - 2y + 4 = 0$

Q. **Find the equations to the common tangents of the parabolas $y^2 = 4ax$ and $x^2 = 4by$.**

Sol. Equation of tangent to $y^2 = 4ax$ is $y = mx + \frac{a}{m}$ ... (i)

If this line also touches parabola $x^2 = 4by$ ... (ii)
then solve (i) and (ii), we get

$x^2 - 4bmx - \frac{4ab}{m} = 0$

Now condition of tangency is $D = 0$.

this gives $m^3 = -\frac{a}{b}$ or $m = -\left(\frac{a}{b}\right)^{1/3}$

equation of common tangent is $y = \left(-\frac{a}{b}\right)^{1/3} x + a\left(-\frac{b}{a}\right)^{1/3}$.

Q. **Let the tangent to the parabola $y^2 = 4ax$ meet the axis in T and the tangent at vertex A in P. If the rectangle TAPQ is completed. Find locus of Q.**

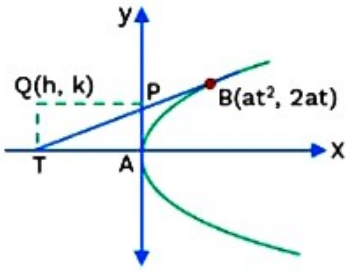
Sol. The tangent at any point $B(at^2, 2at)$ to the parabola is $ty = x + at^2$

Since the tangent at vertex A(0,0) is the y – axis, T and P are $(-at^2, 0)$ and $(0, at)$, respectively.

If Q is (h, k), then $h = -at^2$ and $k = at$.

Eliminating t, we get $k^2 + ah = 0$.

Hence, the locus of Q is $y^2 + ax = 0$



Q. **A pair of tangents are drawn which are equally inclined to a straight line $y = mx + c$ whose inclination to the axis is α, prove that the locus of their point of intersection is the straight line $y = (x - a)\tan 2\alpha$**

Sol. We have $\theta = \alpha_1 - \alpha = \alpha - \alpha_2$

$\therefore \quad 2\alpha = \alpha_1 + \alpha_2$

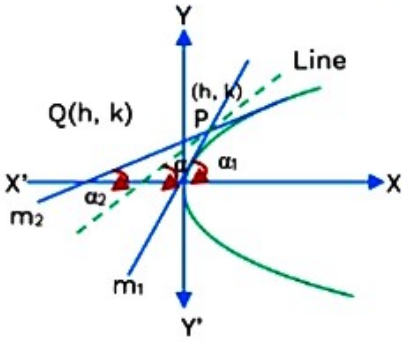
$$\Rightarrow \quad \tan 2\alpha = \tan(\alpha_1 + \alpha_2) = \frac{m_1 + m_2}{1 - m_1 m_2}$$

but (h,k) lies on $y = mx + \frac{a}{m}$

$$\Rightarrow \quad m^2 h - km + a = 0$$

Hence, $m_1 + m_2 = \frac{k}{h}$ and $m_1 m_2 = \frac{a}{h}$

$$\Rightarrow \tan 2\alpha = \frac{\frac{k}{h}}{1 - \frac{a}{h}} = \frac{k}{h - a} \Rightarrow y = (x - a)\tan 2\alpha$$



EQUATION OF NORMALS

1. Point form :

The tangent to the parabola $y^2 = 4ax$ at (x_1, y_1) have equation

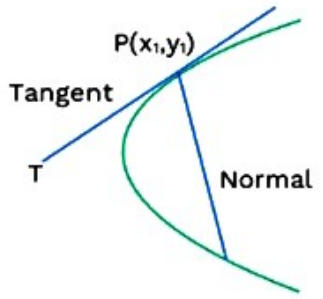
$yy_1 = 2a(x + x_1)$ ... (1)

Therefore, slope of the tangent at $(x_1, y_1) = 2a/y_1$

$\therefore$ Slope of the normal at $(x_1, y_1) = -y_1/2a$

Hence the equation of normal at (x_1, y_1) is

$$y - y_1 = -\frac{y_1}{2a}(x - x_1)$$



2. Slope form :

The equation of normal to the parabola

$y^2 = 4ax$ at (x_1, y_1) is

$$y - y_1 = -\frac{y_1}{2a}(x - x_1) \quad \ldots (1)$$

Let m be the slope of the normal

then $m = -\frac{y_1}{2a}$ or $y_1 = -2am$

$\therefore x_1 = am^2$

Replacing the values of x_1 and y_1 in (1) we get

$y + 2am = m(x - am^2)$ ... (2)

Thus, $y = mx - 2am - am^3$ is a normal to the parabola $y^2 = 4ax$ where m denotes the slope of normal. The coordinates of point of contact are represented by $(am^2, -2am)$.

Hence $y = mx + c$ can be normal to parabola, if and only if

$c = -2am - am^3$.

Equation of parabolas	Point of contact in terms of slope (m)	Equation of normals in terms of slope (m)	Condition of normality
$y^2 = 4ax$	$(am^2, -2am)$	$y = mx - 2am - am^3$	$c = -2am - am^3$
$x^2 = 4ay$	$\left(-\frac{2a}{m}, \frac{a}{m^2}\right)$	$y = mx + 2a + \frac{a}{m^2}$	$c = 2a + \frac{a}{m^2}$

3. Parametric form :

Equation of normal of the parabola $y^2 = 4ax$ at (x_1, y_1) is

$$y - y_1 = -\frac{y_1}{2a}(x - x_1) \quad \ldots (1)$$

Replacing x_1 by at^2 and y_1 by 2 at then (1) becomes

$y - 2at = -t(x - at)^2$

or $y = -tx + 2at + at^3$

Three supplementary Results:

(a) Point of intersection of normal at any two points on the parabola :

Let the points $P\left(at_1^2, 2at_1\right)$ and $Q\left(at_2^2, 2at_2\right)$ lie on the parabola $y^2 = 4ax$

The equations of the normals at $P\left(at_1^2, 2at_1\right)$ and $Q\left(at_2^2, 2at_2\right)$ are

$$y = -t_1x + 2at_1 + at_1^3 \quad \ldots (1)$$

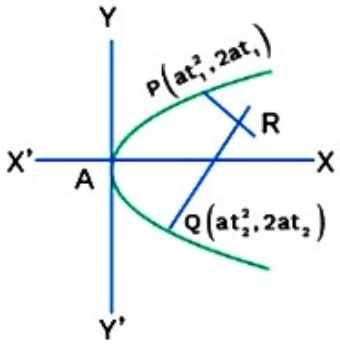
and $y = -t_2x + 2at_2 + at_2^3 \quad \ldots (2)$

Solving (1) and (2), we get point of intersection

$$x = 2a + a\left(t_1^2 + t_2^2 + t_1t_2\right)$$

$$y = -at_1t_2\left(t_1 + t_2\right)$$

If R denotes the point of intersection, then it is $R = \left[2a + a\left(t_1^2 + t_2^2 + t_1t_2\right), -at_1t_2\left(t_1 + t_2\right)\right]$



(b) Relation between t_1 and t_2 if normal at t_1 meets the parabola again at t_2

If the parabola be $y^2 = 4ax$, equation of normal at $P\left(at_1^2, 2at_1\right)$ is

$$y = -t_1x + 2at_1 + at_1^3 \quad \ldots (1)$$

Since normal meet the parabola again at $Q\left(at_2^2, 2at_2\right)$

$$\therefore \quad 2at_2 = -at_1t_2^2 + 2at_1 + at_2^3$$

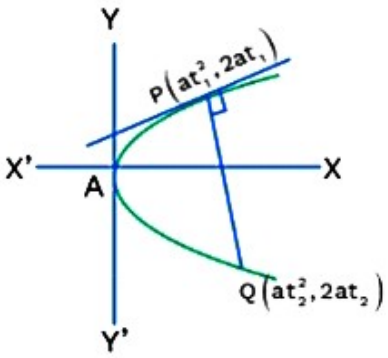
$$\Rightarrow 2a\left(t_2 - t_1\right) + at_1\left(t_2^2 - t_1^2\right) = 0$$

$$\Rightarrow a\left(t_2 - t_1\right)\left[2 + t_1\left(t_2 + t_1\right)\right] = 0$$

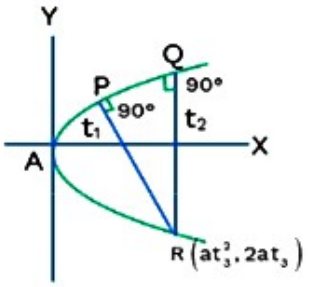
$\because \; a\left(t_2 - t_1\right) \neq 0$

($\because$ t_1 and t_2 are different)

$$\therefore 2 + t_1\left(t_2 + t_1\right) = 0 \quad \therefore \quad t_2 = -t_1 - \frac{2}{t_1}$$



(c) If a normal to the parabola $y^2 = 4ax$ drawn at any point $(at^2, 2at)$ intersects the parabola at t_3 then :



$$t_3 = -t - \frac{2}{t}$$

$\Rightarrow t^2 + tt_3 + 2 = 0$ ... (i)

There are two roots t_1 and t_2. Hence, there are two points $P(t_1)$ and $Q(t_2)$ possible on the parabola from where normals can be drawn to meet parabola at $R(t_3)$

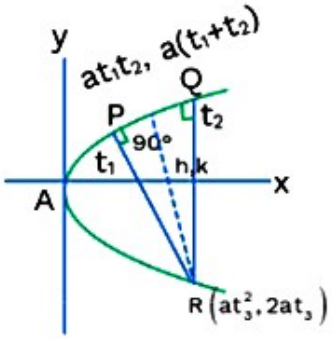
$\Rightarrow t_1 + t_2 = -t_3$ and $t_1t_2 = 2$

Q. **TP and TQ are tangents to the parabola $y^2 = 4ax$ and the normal at P and Q meet at a point R on the curve, show that the centre of the circle which circumscribes the triangle TPQ lies on the parabola $2y^2 = a(x - a)$**

Sol. Normal at two points $P(t_1)$ and $P(t_2)$ meet parabola at $R(t_3)$

$\Rightarrow t_1 + t_2 = -t_3$ & $t_1t_2 = 2$

$\therefore \; T\left(at_1t_2, a\left(t_1 + t_2\right)\right) \equiv T\left(2a, -at_3\right)$



Let centre of circle is (h, k) then it is mid point of T and R.

$\therefore 2h = 2a + at_3^2$ ... (1)

$2k = -at_3 + 2at_3$ ... (2)

i.e. $2k = at_3 \Rightarrow t_3 = \frac{2k}{a}$

Put value of t_3 in equation (i) we get the required locus.

Now replace h by x and k by y

We get $2y^2 = a(x - a)$.

Q. **Find the equation of a normal at the parabola $y^2 = 4x$ which passes through (3, 0).**

Sol. Equation of Normal $y = mx - 2am - am^3$

Here a = 1 and it passes through (3, 0)

$0 = 3m - 2m - m^3$

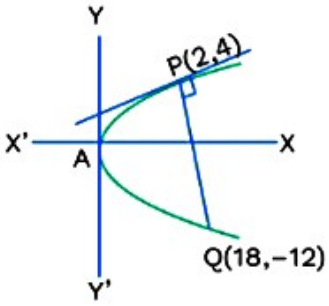
$\Rightarrow m^3 - \; = 0$

$\Rightarrow m = 0, \pm 1$

for m = 0 $\Rightarrow$ y = 0
m = 1 $\Rightarrow$ y = x – 3
m = –1 $\Rightarrow$ y = – x + 3

Q. **Show that normal to the parabola $y^2 = 8x$ at the point (2, 4) meets it again at (18, – 12). Also find the length of the normal chord.**

Sol. Comparing the given parabola with $y^2 = 4ax$.
$\therefore$ 4a = 8
$\therefore$ a = 2
$P\left(at_1^2, 2at_1\right) \equiv (2, 4)$
$\Rightarrow t_1 = 1$
$\Rightarrow$ parameter at $Q(t_2) = -t_1 - \frac{2}{t_1} = -3$
$\therefore\ Q\left(2(-3)^2, 2\times 2(-3)\right)$



i.e. Q(18, – 12)
$\therefore$ Length of normal chord PQ = Distance between points P and Q
$= PQ = \sqrt{(18-2)^2 + (-12-4)^2} = 16\sqrt{2}$

Q. **Normal to the parabola $y^2 = 12x$ at P(3, 6) meet it again at the point Q. Find the equation of the circle described on PQ as diameter.**

Sol. a = 3
$2at_1 = 6$
$t_1 = 1$
P(3, 6)
$t_2 = -t_1 - \frac{2}{t_1}$

$t_2 = -3$
Q will be (27, – 18)
equation of circle
(x – 3)(x – 27) + (y – 6)(y + 18) = 0

Q. **If the normal to a parabola $y^2 = 4ax$, makes an angle ϕ with the axis show that it will again cut the curve at an angle $\tan^{-1}\left(\frac{1}{2}\tan\phi\right)$**

Sol. Let the normal at $P\left(at_1^2, 2at_1\right)$ be $y = -t_1x + 2at_1 + at_1^3$

$\therefore$ $\tan\phi = -t_1$ = slope of the normal ... (1)

It meet the curve again Q say $\left(at_2^2, 2at_2\right)$

$$t_2 = -t_1 - \frac{2}{t_1} \quad \text{... (2)}$$

$\therefore t_1 = 2$ from equation (1), $t_2 = -3$

Now angle between the normal and parabola = Angle between the normal and tangent at

$Q\left(\right.$ i.e., $t_2y = x + at_2^2\left.\right)$

If θ be the angle, then

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1m_2} = \frac{-t_1 - \frac{1}{t_2}}{1 + \left(-t_1\right)\left(\frac{1}{t_2}\right)} = -\frac{t_1t_2 + 1}{t_2 - t_1} = \frac{t_1\left(-t_1 - \frac{2}{t_1}\right) + 1}{-t_1 - \frac{2}{t_1} - t_1}$$

{from equation (1)}

$$\theta = \tan^{-1}\left(\frac{1}{2}\tan\phi\right)$$

Q. **Prove that the normal chord to a parabola $y^2 = 4ax$ at the point whose ordinate is equal to abscissa subtends a right angle at the focus.**

Sol. let the normal $P\left(at_1^2, 2at_1\right)$ meet the curve at $Q(at_2^2, 2at_2)$

$t_1 = 2$ & $t_2 = -t_1 - 2/t_1 = -3$

$\therefore$ PQ is a normal chord.

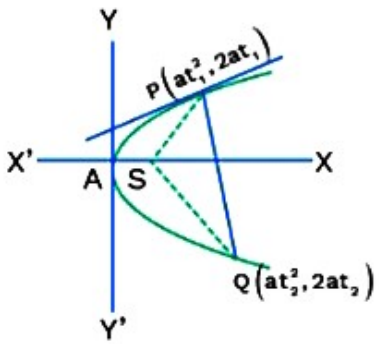
$$\therefore \quad \text{Slope of SP} = \frac{4a - 0}{4a - a} = \frac{4a}{3a} = \frac{4}{3}$$

$$\text{and slope of SQ} = = \frac{-6a - 0}{9a - a} = \frac{6a}{8a} = -\frac{3}{4}$$

$$\text{Now slope of SP} \times \text{slope of SQ} = \frac{4}{3} \times -\frac{3}{4} = -1$$

$$\therefore \angle PSQ = \frac{\pi}{2} \text{ i.e. PQ subtends a right angle}$$

at the focus S.

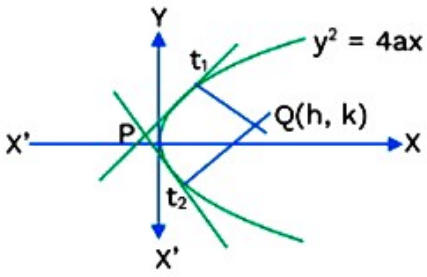


Q. **If tangents are drawn to $y^2 = 4ax$ from any point P on the parabola $y^2 = a(x + b)$ then show that the normals drawn at their point of contact meet on a fixed line.**

Sol. Let $P(at_1t_2, a(t_1 + t_2))$; P must satisfy $y^2 = a(x + b)$

Hence $a^2(t_1 + t_2)^2 = a((at_1t_2) + b)$

$a\left(t_1^2 + t_2^2 + t_1t_2\right) = b$



Now coordinates of point of intersection of normals at t_1 and t_2 are

$h = a\left(t_1^2 + t_2^2 + t_1t_2 + 2\right)$ ... (1)

and $k = -at_1t_2(t_1 + t_2)$ ... (2)

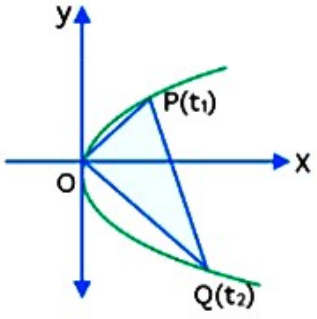
From Eq. (i), $h = b + 2a$

$\Rightarrow x = b + 2a$

Q. **If a chord normal to the parabola $y^2 = 4ax$ at one end, subtends a right angle at the vertex, show that it is inclined at an angle $\tan^{-1}\sqrt{2}$ to the axis and the normal chord passes through (4a, 0).**

Sol. Let $y^2 = 4ax$ be the parabola and the normal drawn to it at $P(t_1)$ meets the parabola again at $Q(t_2)$ and PQ subtends 90° at the vertex O. We have

$t_2 = -t_1 - \frac{2}{t_1}$



Slope of $OP = \frac{2at_1}{at_1^2} = \frac{2}{t_1}$

Slope of $OQ = \frac{2}{t_2}$

OP and OQ are perpendicular, therefore $\frac{2}{t_1} \cdot \frac{2}{t_2} = -1$

$\Rightarrow \quad 4 = -t_1t_2 = -\left(-t_1^2 - 2\right)$

[From the equation (1) $t_1t_2 = -t_1^2 - 2$] or

$t_1^2 = 2 \Rightarrow t_1 = \pm\sqrt{2}$

Now the slope of the normal at $P(t_1) = -t_1 = \mp\sqrt{2}$

$\therefore$ The normal chord is inclined at an angle $\tan^{-1}(\sqrt{2})$ to the axis.

Q. **Prove that two parabola $y^2 = 4ax$ and $y^2 = 4c(x - b)$ cannot have a common normal other than the axis unless $\frac{b}{a-c} > 2$.**

Sol. Equation of any normal to parabola $y^2 = 4ax$ is

$y = m_1x - 2am_1 - am_1^3$ ... (1)

Equation of any normal to parabola $y^2 = 4c(x - b)$ is

$y = m_2(x - b) - 2cm_2 - cm_2^3$ ... (2)

If the equations (1) and (2) represent the same line, then

$m_1 = m_2 = m$ (say) ... (3)

and $-2am_1 - am_1^3 = -bm_2 - 2cm_2 - cm_2^3$

$\Rightarrow 2am + am^3 = bm + 2cm + cm^3$ [From the eq. (3)]

or $m\left[(a - c)m^2 + (2a - 2c - b)\right] = 0$

i.e. $m = 0$ or $m^2 = \frac{b - 2(a - c)}{a - c}$ If $m = 0$, then axis is the common normal.

For the common normal other than the axis, the values of m obtained from the other equation must be real and different from zero. Thus, we have

$m^2 > 0 \Rightarrow \frac{b - 2(a - c)}{a - c} > 0 \Rightarrow \frac{b}{(a - c)} > 2$, Which is the desired result.

Q. **If $a^2 > 8b^2$, prove that a point can be formed such that two tangents drawn from it to the parabola $y^2 = 4ax$ are normals to the parabola $x^2 = 4by$.**

Sol. Equation of a tangent to parabola $y^2 = 4ax$ is

$$y = mx + \frac{a}{m} \quad \text{... (1)}$$

Equation of a normal to parabola $x^2 = 4by$ is

$$x = My - 2bM - bM^3$$

$$\Rightarrow \quad y = \frac{1}{M}x + 2b + bM^2 \quad \text{... (2)}$$

If the lines (1) and (2) are same, then

$$\frac{1}{M} = m \text{ and } 2b + bM^2 = \frac{a}{m}$$

$$\Rightarrow \quad 2b + \frac{b}{m^2} = \frac{a}{m} \Rightarrow 2bm^2 - am + b = 0$$

Above equation is a quadratic in 'm' it has real and distinct roots if its discriminant is positive $\therefore (-a)^2 - 4(2b)\, b > 0 \Rightarrow a^2 > 8b^2$
Which is the desired result.

Q. **Let P be the point on the parabola $y^2 = 4x$ which is at the shortest distance from the center S of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$. Let Q be the point on the circle dividing the line segment SP internally. Then**

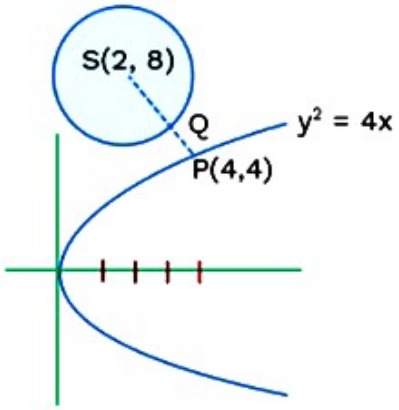
(A) $SP = 2\sqrt{5}$

(B) $SQ : QP = (\sqrt{5} + 1) : 2$

(C) The x-intercept of the normal to the parabola at P is 6

(D) The slope of the tangent to the circle at Q is $\frac{1}{2}$

Sol. **(ACD)**



Given circle is :

$x^2 + y^2 - 4x - 16y + 64 = 0$

or $(x - 2)^2 + (y - 8)^2 = 4$

Centre $S \equiv (2, 8)$

normal to parabola is $y = mx - 2m - m^3$

For shortest distance from the centre, normal must pass through the centre of the circle

$\Rightarrow \quad 8 = 2m - 2m - m^3$

$\Rightarrow \quad m = -2$

$\Rightarrow$ Normal at P, $y = -2x + 12$

$\therefore$ Point $P \equiv (am^2, -2am) \equiv (4, 4)$

$$\therefore \quad SP = \sqrt{(4-2)^2 + (8-4)^2} = 2\sqrt{5}$$

$\therefore \quad SQ : QP = 2 : (2\sqrt{5} - 2)$

$$= \frac{2}{(2\sqrt{5} - 2)} \frac{(\sqrt{5} + 1)}{(\sqrt{5} + 1)} = \frac{(\sqrt{5} + 1)}{5 - 1} = \frac{\sqrt{5} + 1}{4}$$

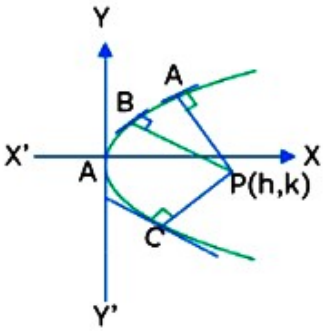
Slope of tangent at $Q = \frac{1}{2}$

(Tangent at Q is perpendicular to SP)

CO-NORMAL POINTS :

Maximum three normal can be drawn from a point to a parabola and their feet (points where the normal meet the parabola) are called co-normal points.

Let P(h, k) be any given point and $y^2 = 4ax$ be a parabola.



The equation of any normal to $y^2 = 4ax$ is

$y = mx - 2am - am^2$

If it passes through (h, k) then

$k = mh - 2am - am^3$

$\Rightarrow am^3 + m(2a - h) + k = 0$ ... (i)

This is a cubic equation in m, so it has three roots, say

m_1, m_2 and m_3

$\therefore m_1 + m_2 + m_3 = 0$ ... (ii)

$m_1m_2 + m_2m_3 + m_3m_1 = \frac{(2a - h)}{a}$ ... (iii)

$m_1m_2m_3 = \frac{-k}{a}$ ... (iv)

Hence for any given point P(h, k), (i) has three real or imaginary roots. Corresponding to each of these three roots, we have each normal passing through P(h, k). Hence we have three normals PA, PB and PC drawn through P to the parabola.

Points A, B, C in which the three normals from P(h, k) meet the parabola are called co-normal points.

Properties of co-normal points :

(1) The algebraic sum of the slopes of three concurrent normals is zero. This follows from equation (ii).

(2) The algebraic sum of ordinates of the feet's of three normals drawn to a parabola from a given point is zero.
Let the ordinates of A,B,C be y_1, y_2, y_3 respectively then
$y_1 = -2am_1$, $y_2 = -2am_2$ and $y_3 = -2am_3$
$\therefore$ Algebraic sum of these ordinates is

$$\begin{aligned} y_1 + y_2 + y_3 &= -2am_1 - 2am_2 - 2am_3 \\ &= -2a(m_1 + m_2 + m_3) \\ &= -2a \times 0 \text{ \{From equation (ii)\}} \\ &= 0 \end{aligned}$$

(3) If three normals drawn to any parabola $y^2 = 4ax$ from a given point (h,k) is real then $h > 2a$. When normals are real, then all the three roots of equation (i) are real and in that case

$m_1^2 + m_2^2 + m_3^2 > 0$ (for any values of m_1, m_2, m_3)

$\Rightarrow (m_1 + m_2 + m_3)^2 - 2(m_1m_2 + m_2m_2 + m_3m_1) > 0$

$\Rightarrow (0)^2 - \frac{2(2a-h)}{a} > 0$

$\Rightarrow h - 2a > 0$
or $h > 2a > 0$

(4) The centroid of the triangle formed by the feet of the three normals lies on the axis of the parabola. If $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ be vertices of ΔABC, then its centroid is

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right) = \left(\frac{x_1 + x_2 + x_3}{3}, 0\right)$$

Since $y_1 + y_2 + y_3 = 0$ (from result-2). Hence the centroid lies on the x - axis, which is the axis of the parabola also.

Now $\frac{x_1 + x_2 + x_3}{3} = \frac{1}{3}\left(am_1^2 + am_2^2 + am_3^2\right) = \frac{a}{3}\left(m_1^2 + m_2^2 + m_3^2\right)$

$= \frac{a}{3}\left\{(m_1 + m_2 + m_3)^2 - 2(m_1m_2 + m_2m_3 + m_3m_1)\right\}$

$= \frac{a}{3}\left\{(0)^2 - 2\left\{\frac{2a-h}{a}\right\}\right\} = \frac{2h - 4a}{3}$

$\therefore$ Centroid of ΔABC is $\left(\frac{2h - 4a}{3}, 0\right)$

Q. **If from point P(h, k) three normals are drawn to the parabola $(y - 1)^2 = 8(x - 2)$ then find the condition**

Sol. Here $a = 2$ and abscissa of point from where three normals are drawn must be greater than $2a$ i.e. $x - 2 > 2a$ i.e.
$x > 6$ Hence $h > 6$

Q. **The ordinate of points P and Q on the parabola $y^2 = 12x$ are in the ratio of 1 : 2 Find the locus of the point of intersection of the normal to the parabola at P and Q.**

Sol. Here $a = 3$,
Let $P\left(3t_1^2, 6t_1\right)$ and $Q\left(3t_2^2, 6t_2\right)$ lie on the parabola.

According to the question, $\frac{6t_1}{6t_2} = \frac{1}{2} \Rightarrow t_2 = 2t_1$ ... (i)

Let $P(\alpha,\beta)$ be the point of intersection of normal to parabola at P and Q then

$\alpha = 2a + a\left(t_1^2 + t_2^2 + t_1t_2\right) = 6 + 21t_1^2$ ... (ii)

and $\beta = -at_1t_2\left(t_1 + t_2\right) = -18t_1^3$ ... (iii)

from (ii) $t_1^6 = \left(\frac{\alpha - 6}{21}\right)^3$ and from (iii)

$t_1^6 = \frac{\beta^2}{324}$

equate t_1^6, we get

$343\beta^2 = 12(\alpha - 6)^3$

$\therefore$ Locus is $343y^2 = 12(x - 6)^3$

Q. **Find the locus of points through which three normals to parabola $y^2 = 4ax$ passes and two of them are perpendicular to each other.**

Sol. Let P(h, k) be the point of intersection of three normals to be parabola $y^2 = 4ax$ then it will be

$k = mh - 2am - am^3$

$y = mx - 2am - am^3$ and it passes through P(h, k)

$\therefore am^3 + m(2a - h) + k = 0$ ... (i)

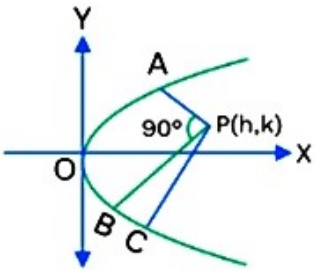
It has three roots m_1, m_2 and m_3

$m_1 + m_2 + m_3 = 0$ ... (ii)

$m_1m_2m_3 = \frac{-k}{a}$ ... (iii)

Given condition, $m_1m_2 = -1$ ... (iv)

as given in figure



from (iii) and (iv)

$m_3 = \frac{k}{a}$

$\because m_3$ is also a root of equation (1) therefore it will satisfy equation (i)

$a\left(\frac{k}{a}\right)^3 + (2a - h)\left(\frac{k}{a}\right) + k = 0$

$\Rightarrow k^3 + (2a - h)ka + ka^2 = 0$

$\Rightarrow k^2 + 3a^2 - ah = 0$

$\therefore$ locus of P(h, k) is

$y^2 + 3a^2 - ax = 0$

Q. **A circle circumscribing the triangle formed by three co-normal points passes through the vertex of the parabola and its equation is**
$2(x^2 + y^2) - 2(h + 2a)x - ky = 0$

Sol. Consider a circle

$x^2 + y^2 + 2gx + 2fy + c = 0$ ...(1)

If the circle (1) passes through feet of normals Q_1, Q_2 and Q_3 then coordinates $(am^2, -2am)$ must satisfy the equation of the circle, (where $m = m_1, m_2, m_3$). So

$(am^2)^2 + (-2am)^2 + 2g(am^2) + 2f(-2am) + c = 0$

or $a^2m^4 + (4a^2 + 2ag)m^2 - 4afm + c = 0$ ... (2)

This is a biquadratic equation in the variable 'm'. Its roots are m_1, m_2, m_3 and m_4 first three of which are the slopes of three normals at Q_1, Q_2 and Q_3.

Now, $m_1 + m_2 + m_3 + m_4 = 0$

$\Rightarrow \; m_4 = 0$

Thus, the circle passes through the points

$\left(am_2^2, -2am_2\right), \left(am_3^2, -2am_3\right)$ and $\left(am_4^2, -2am_4\right)$, where $m_4 = 0$

The fourth point is $(0,0)$ and therefore $c = 0$. The equation (2) now becomes

$$a^2m^4 + \left(4a^2 + 2ag\right)m^2 - 4amf = 0$$

$\Rightarrow \; am^3 + (4a + 2g)m - 4f = 0$ ... (3)

Equation (3) must be identical. We have

$4a = 2g = 2a - h$

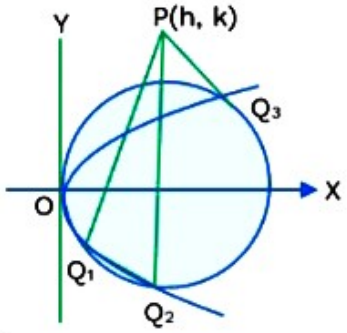
and $-4f = k$

$\Rightarrow 2g = -(2a + h)$

and $2f = -k/2$

The equation (1) then becomes.

$$x^2 + y^2 - (2a + h)x - \frac{k}{2}y = 0 \quad \text{... (4)}$$

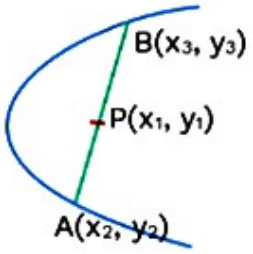


Above equation (4) represents the equation of the circle passing through the feet of normals Q_1, Q_2 and Q_3 drawn from a point P(h, k) to the parabola $y^2 = 4ax$.

CHORD OF THE PARABOLA {WHOSE MIDDLE POINT IS GIVEN} :

Equation of the parabola is $y^2 = 4ax$ ... (1)

Let AB be a chord of the parabola whose middle point is $P(x_1, y_1)$.



Equation of chord AB is

$y - y_1 = m(x - x_1)$ ... (2)

where m = slope of AB

Let $A = (x_2, y_2)$ and $B = (x_3, y_3)$

Since A and B lie on parabola (1)

$\therefore y_2^2 = 4ax_2$ and $y_3^2 = 4ax_3$

$$\therefore y_2^2 - y_3^2 = 4a(x_2 - x_3) \text{ or } \frac{y_2 - y_3}{x_2 - x_3} = \frac{4a}{y_2 + y_3} \quad \text{... (3)}$$

But $P(x_1, y_1)$ is the middle point of AB

$y_2 + y_3 = 2y_1$

$$\therefore \text{ From (3), } \frac{y_2 - y_3}{x_2 - x_3} = \frac{4a}{2y_1} = \frac{2a}{y_1}$$

$$\therefore \text{ Slope of AB i.e., } m = \frac{2a}{y_1} \quad \text{.... (4)}$$

From (2), equation of chord AB is $y - y_1 = \frac{2a}{y_1}(x - x_1)$

or

$yy_1 - y_1^2 = 2ax - 2ax_1$ or $yy_1 - 2ax = y_1^2 - 2ax_1$

or $yy_1 - 2a(x - x_1) = y_1^2 - 4ax_1$ [Subtracting $2ax_1$ from both sides] ... (5)

(5) is the required equation. In usual notations, equation (5) can be written as $T = S_1$.

The same result holds true for circle, ellipse and hyperbola also.

Q. **Find the locus of the mid points of the chords of the parabola $y^2 = 4ax$ which subtend a right angle at the vertex of the parabola.**

Sol. Let P(h,k) be the mid point of a chord QR of the parabola $y^2 = 4ax$ then equation of the chord QR is $T = S_1$

or $yk - 2a(x + h) = k^2 - 4ah$

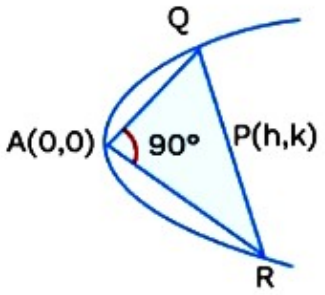
$\Rightarrow \quad yk - 2ax = k^2 - 2ah \qquad \ldots (1)$

If A is the vertex of the parabola. For combined equation of AQ and AR making homogeneous of $y^2 = 4ax$ with the help of (1)

$y^2 = 4ax(1)$

$$y^2 = 4ax\left(\frac{yk - 2ax}{k^2 - 2ah}\right)$$

$$y^2(k^2 - 2ah) - 4akxy + 8a^2x^2 = 0$$



Since $\angle QAF = 90^\circ$

$\therefore$ coefficient of x^2 + coefficient of $y^2 = 0$

$k^2 - 2ah + 8a^2 = 0$

Hence the locus of P(h, k) is $y^2 - 2ax + 8a^2 = 0$

Q. **If a chord, which is not a tangent, of the parabola $y^2 = 16$ has the equation $2x + y = p$, and midpoint (h, k), then which of the following is(are) possible value(s) of p, h and k ?**

(A) $p = -1, h = 1, k = -3$ **(B) $p = 2, h = 3, k = -4$**

(C) $p = -2, h = 2, k = -4$ **(D) $p = 5, h = 4, k = -3$**

Sol. **(D)**

Parabola, $y^2 = 16x$

Equation of chord, $2x + y = p \qquad \ldots (1)$

Equation of chord having mid point (h,k) is $T = S_1$

$ky - 8(x + h) = k^2 - 16h \qquad \ldots (2)$

or $ky - 8x = k^2 - 8h$

Comparing (1) and (2), we get

$$\frac{k}{1} = \frac{-8}{2} = \frac{k^2 - 8h}{p}$$

$\Rightarrow k = -4$ and $k^2 - 8h = -4p$

From this, we get

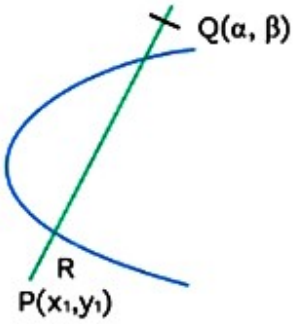
$16 - 8h = -4p$

Clearly, $h = 3$ and $p = 2$.

PAIR OF TANGENTS

Let the parabola be $y^2 = 4ax$ ... (1)

Let $P(x_1, y_1)$ be a point outside the parabola.



Let a chord of the parabola through the point $P(x_1, y_1)$ cut the parabola at R and let $Q(\alpha,\beta)$ be an arbitrary point on line PR. Let R divide PQ in the ratio $\lambda : 1$,

then $R = \left(\frac{\lambda\alpha + x_1}{\lambda+1}, \frac{\lambda\beta + y_1}{\lambda+1}\right)$

Since lies on parabola (1), therefore,

$$\left(\frac{\lambda\beta + y_1}{\lambda+1}\right)^2 - 4a\left(\frac{\lambda\alpha + x_1}{\lambda+1}\right) = 0$$

or $(\lambda\beta + y_1)^2 - 4a(\lambda\alpha + x_1)(\lambda + 1) = 0$

or

$$(\beta^2 - 4a\alpha)\lambda^2 + 2[\beta y_1 - 2a(\alpha + x_1)]\lambda + (y_1^2 - 4ax_1) = 0$$

... (2)

Line PQ will become tangent to parabola (1) if roots of equation (2) are equal or if

$$4[\beta y_1 - 2a(\alpha + x_1)]^2 = 4(\beta^2 - 4a\alpha)(y_1^2 - 4ax_1)$$

Hence, locus of $Q(\alpha, \beta)$ i.e. equation of pair of tangents from $P(x_1, y_1)$ is

$$[yy_1 - 2a(x + x_1)]^2 = (y^2 - 4ax)(y_1^2 - 4ax_1)$$

$\Rightarrow SS_1 = T^2$

where S, S_1 and T have usual meanings.

The same result holds true for cicle ellipse and hyperbola also.

Q. **Tangents are drawn from the point (-1, 2) on the parabola $y^2 = 4x$. The length, these tangents will intercept on the line $x = 2$ is:**

(A) 6 **(B)** $6\sqrt{2}$ **(C)** $2\sqrt{6}$ **(D) none of these**

Sol. **(B)**

$SS_1 = T^2$

$$(y^2 - 4x)(y_1^2 - 4x_1) = (yy_1 - 2(x + x_y))^2$$

$$(y^2 - 4x)(4 + 4) = [2y - 2(x - 1)]^2 = 4(y - x + 1)^2$$

$$2(y^2 - 4x) = (y - x + 1)^2$$

solving with the line $x = 2$ we get,

$2(y^2 - 8) = (y - 1)^2$ or $2(y^2 - 8) = y^2 - 2y + 1$

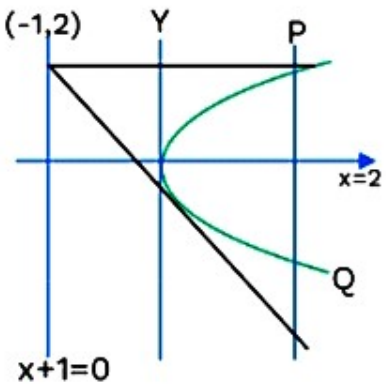
or $y^2 + 2y - 17 = 0$

where $y_1 + y_2 = -2$ and $y_1y_2 = -17$

Now $|y_1 - y_2|^2 = (y_1 + y_2)^2 - 4y_1y_2$

or $|y_1 - y_2|^2 = 4 - 4(-17) = 72$

$\therefore (y_1 - y_2) = \sqrt{72} = 6\sqrt{2}$



CHORD OF CONTACT OF POINT WITH RESPECT TO A PARABOLA

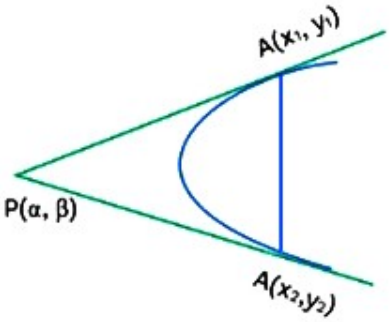
Two tangents PA and PB are drawn to parabola, then line joining AB is called the chord of contact to the parabola with respect to point P.

Let the parabola be $y^2 = 4ax$ ... (1)

Let $P(\alpha,\beta)$ be a point outside the parabola.

Let PA and PB be the two tangents from $P(\alpha,\beta)$ to parabola (1).

Let $A = (x_1, y_1)$ and $B = (x_2, y_2)$



Equation of the tangent PA is

$yy_1 = 2a(x + x_1)$ ... (2)

Equation of the tangent PB is

$yy_2 = 2a(x + x_2)$ ... (3)

Since lines (2) and (3) pass through $P(\alpha,\beta)$, therefore

$\beta y_1 = 2a(a + x_1)$ ... (4)

and $\beta y_2 = 2a(a + x_2)$ ... (5)

Now we consider the equation

$y\beta = 2a(x + \alpha)$ ... (6)

From (4) and (5), it follows that line (6) passes through $A(x_1, y_1)$ and $B(x_2, y_2)$.

Hence (6) is the equation of line AB which is the chord of contact of point $P(\alpha,\beta)$ with

respect to parabola (1) i.e, chord of contact is $y\beta = 2a(x + \alpha)$

The same result holds true for circle, ellipse and hyperbola also.

Q. **Tangent are drawn to parabola $y^2 = 4ax$ at point where the line $lx + my + n = 0$ meets the parabola. Find the point of intersection of these tangents.**

Sol. Let the tangent intersect at P(h, k), then $lx + my + n = 0$ will be the chord of contact of P. That means $\ell x + my + n = 0$ and $yk - 2ax - 2ah = 0$ will represent the same line. Thus,

$$\frac{k}{m} = \frac{-2a}{l} = \frac{-2ah}{n} \Rightarrow h = \frac{n}{l}, k = -\frac{2am}{l}$$

thus $\left(\frac{n}{\ell}, \frac{-2am}{\ell}\right)$ is point of intersection

Q. **Line $3x + y = 6$ intersect the parabola $y^2 = -4x$ at A and B. Find coordinates of point of intersection of Tangent drawn at A and B.**

Sol. Let point of intersection is P(h, k)

equation of chord of contact w.r.t P is

$yk = -2(x + h)$

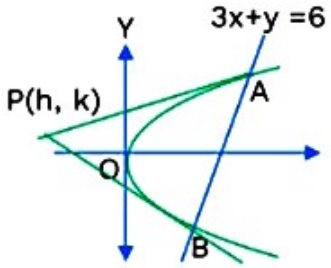
$yk + 2x + 2h = 0$ ... (1)

$3x + y - 6 = 0$ ... (2)

(1) and (2) are same equation comparing

$$\frac{3}{2} = \frac{1}{k} = \frac{-6}{2h}$$

$\Rightarrow k = 2/3 ; h = -2$



Q. **Pair of tangents are drawn to parabola $y^2 = -4x$ from every point on the line $3x + y = 2$. Prove that there chord of contact passes through a fix point.**

Sol. Let point on the line $3x + y = 2$

$(\lambda, 2 - 3\lambda)$

equation of chord of contact be

$y(2 - 3\lambda) = -2(x + \lambda)$

$2y - 3\lambda y = -2x - 2\lambda$

$2y + 2x + \lambda(2 - 3y) = 0$

Solving family of lines,
we get $y = 2/3$
$x = -2/3$

Q. **From a point on the line $x + 4a = 0$ pair of tangents are drawn to the parabola $y^2 = 4x$ Prove that the chord of contact subtends 90° at vertex.**

Sol. Let $P\left(at_1^2, 2at_1\right)$ and $Q\left(at_2^2, 2at_2\right)$ be the points of contact for tangents drawn from any point on the line $x + 4a = 0$. Their point of intersection will be on this line.

$\therefore\ at_1t_2 + 4a = 0$

or $t_1t_2 = -4$

Let O is vertex

Slope of OP $= 2/t_2$; Slope of OO $= 2/t_1$

Now $\frac{2}{t_2} \times \frac{2}{t_1} = -1$ H.P.

This is also the condition for chord PQ to subtend a right angle at the vertex.

Q. **Find locus of mid-point of chord of parabola $y^2 = 4ax$. Which always passes through focus.**

Sol. Any chord PQ which get bisected at point R(h, k) is $T = S_1$
or $ky - 2a(x + h) = k^2 - 4ah$
Now given that this chord is focal chord, then it must passes through focus S(a, 0).
Then $k(0) - 2a(a + h) = k^2 - 4ah$
$\Rightarrow k^2 = 2ah - 4a^2 \Rightarrow k^2 = 2ah - 4a^2$
$\Rightarrow y^2 = 2a(x - a)$

Q. **Prove that the area of the triangle formed by the tangents drawn from (x_1, y_1) to $y^2 = 4ax$ and their chord of contact is $\left(y_1^2 - 4ax_1\right)^{3/2} / 2a$**

Sol. Equation of QR (chord of contact) is

$yy_1 = 2a(x + x_1)$

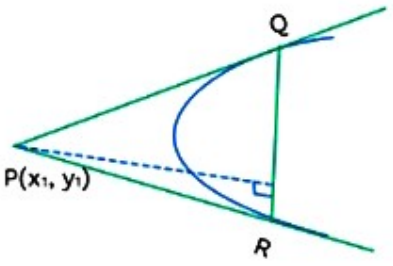
$yy_1 - 2a(x + x_1) = 0$

$\because$ PM = Length of perpendicular from $P(x_1, y_1)$ on QR

$$= \frac{|y_1y_1 - 2a(x_1 + x_1)|}{\sqrt{(y_1^2 + 4a^2)}} = \frac{|(y_1^2 - 4ax_1)|}{\sqrt{(y_1^2 + 4a^2)}}$$

[Since $P(x_1, y_1)$ lies outside the parabola

$\therefore y_1^2 - 4ax_1 > 0$]



Now area of $\Delta PQR = \frac{1}{2} QR \cdot PM$

$$= \frac{1}{2}\frac{1}{|a|}\sqrt{(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}\frac{(y_1^2 - 4ax_1)}{\sqrt{y_1^2 + 4a^2}}$$

$$= (y_1^2 - 4ax)^{3/2} / 2a, \text{ if } a > 0$$

Know the facts

Length of the chord of contact

$$QR = \frac{\sqrt{y_1^2 + 4a^2}\sqrt{y_1^2 - 4ax_1}}{a}$$

DIAMETER OF A PARABOLA

Diameter of a conic is the locus of middle points of a series of its parallel chords.

Equation of diameter of a parabola:

Let the parabola be $y^2 = 4ax$ ...(1)

Let AB be one of the chords of a series of parallel chords having slope m.

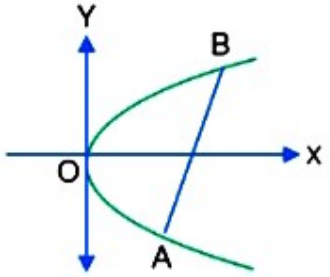
Let $P(\alpha, \beta)$ be the middle point of chord AB, then equation of AB will be $T = S_1$

or $y\beta - 2a(x + \alpha) = \beta^2 - 4\alpha a$ ... (2)

Slope of line $(2) = \frac{2a}{\beta}$

but slope of line (1) i.e. line AB is m.

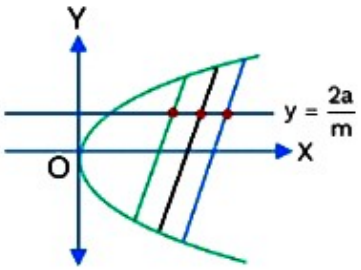
$\therefore \frac{2a}{\beta} = m$ or $\beta = \frac{2a}{m}$



Hence locus of $P(\alpha, \beta)$ i.e. equation of

diameter (which is the locus of a series of a parallel chords having slope m) is $y = \frac{2a}{m}$

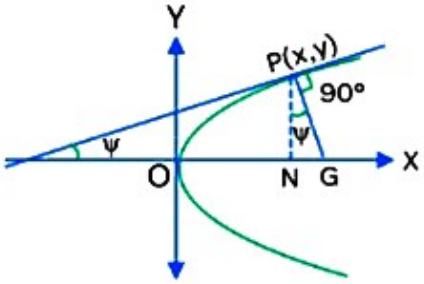
Clearly line (3) is parallel to the axis of the parabola. Thus a diameter of a parabola is parallel to its axis.



Length of tangent, subtangent, normal and sub-normal :

Let the parabola is $y^2 = 4ax$ Let the tangent at any point $P(x,y)$ meet the axis of parabola at T and G respectively and tangent makes an angle ψ with x -axis.

$$\therefore \tan\psi = \left(\frac{dy}{dx}\right)_{P(x,y)} \quad \text{and } PN = y$$



$\therefore$ PT = length of tangent
$= PN\operatorname{cosec}\psi = y\operatorname{cosec}\psi$

PG = length of normal $= y\sec\psi$

TN = length of sub-tangent $= PN\cot\psi = y\cot\psi$

NG = length of sub-normal $= y\tan\psi$

Q. **Find the length of tangent, sub-tangent, normal and sub-normal to $y^2 = 4ax$ at $(at^2, 2at)$.**

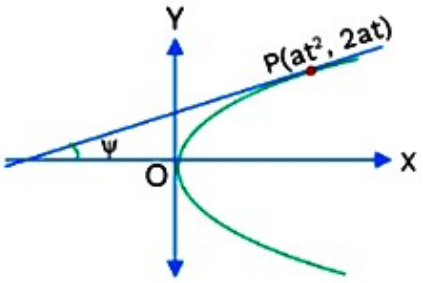
Sol. $y^2 = 4ax$

$$l(\text{normal}) = y\sec\psi = 2at\sqrt{1+\tan^2\psi} = 2a\sqrt{1+t^2}$$

$$l(\text{sub-tangent}) = y\cot\psi = 2at \cdot t = 2at^2$$

$$l(\text{sub-normal}) = y\tan\psi = 2at \cdot \frac{1}{t} = 2a$$

Thus length of sub-normal of parabola is constant and equal to semi-latus rectum.



PROPERTIES OF PARABOLA :

(1) Circle described on the focal length (distance) as diameter touches the tangent at the vertex.

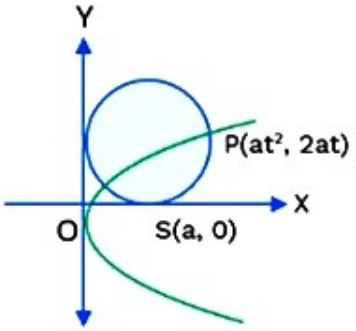
Equation of the circle described on SP as diameter is

$$(x - at^2)(x - a) + (y - 2at)(y - 0) = 0$$

Solve it with y -axis i.e. $x = 0$, we get

$$y^2 - 2aty + a^2t^2 = 0 \Rightarrow (y - at)^2 = 0$$

circle touches y-axis at (0, at)



(2) Circle described on the focal chord as diameter touches directrix

Equation of the circle described on PQ as diameter is

$$(x - at^2)\left(x - \frac{a}{t^2}\right) + (y - 2at)\left(y + \frac{2a}{t}\right) = 0$$

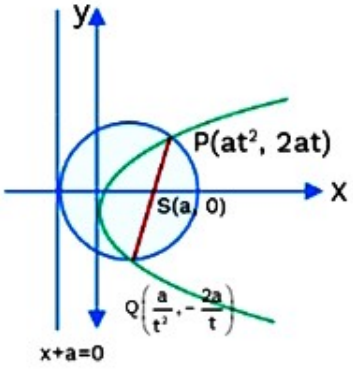
Solving it with $x = -a$

$$\left(-a - at^2\right)\left(-a - \frac{a}{t^2}\right) + (y - 2at)\left(y + \frac{2a}{t}\right) = 0$$

$$\Rightarrow y^2 - 2a\left(t - \frac{1}{t}\right)y + a^2\left(t - \frac{1}{t}\right)^2 = 0$$

$$\Rightarrow \left[y - a\left(t - \frac{1}{t}\right)\right]^2 = 0$$

circle touches the directrix.



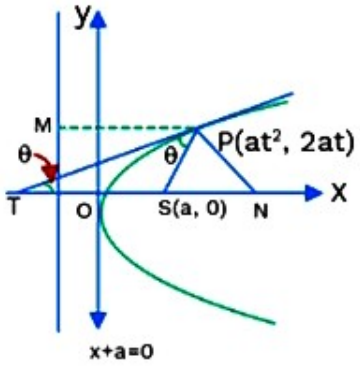
(3) Tangent at P is

$yt = x + at^2$, meet x -axis at T, then $T(-at^2, 0)$

Normal at P is $y + xt = 2at + at^3$, meet x -axis at N

then $N(2a + at^2, 0)$

$\Rightarrow$ $ST = SN = a + at^2 = PM = PS$



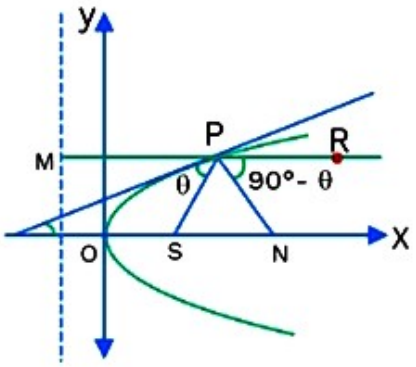
$\Rightarrow \quad \angle PTS = \angle TPS = \theta$

$\because \quad TS = PS = PM \Rightarrow \angle TPM = \theta$

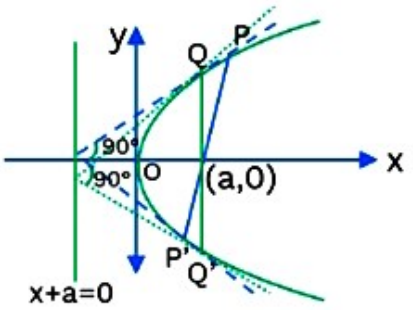
Tangent and Normal at any point P bisect the angle between PS and PM internally and externally. This property leads to the reflection property of parabola.

Circle circumscribing the triangle formed by any tangent normal and x - axis, has its centre at focus. If we extend MP, then from figure $\angle RPN = \angle SPN = 90 - \theta$

Thus ray parallel to axis meet parabola at P and after reflection from P it passes through the focus.



(4) The tanents at the extremities of a focal chord intersect at right angles on the directrix.



(5) The portion of tangent to the parabola intercepted between the directrix and the curve subtends a right angle at the focus

Tangent at $P(at^2, 2at)$ is $yt = x + at^2$ meet the directrix at $x = -a \Rightarrow Q\left(-a, \frac{at^2 - a}{t}\right)$ and S(a, 0).

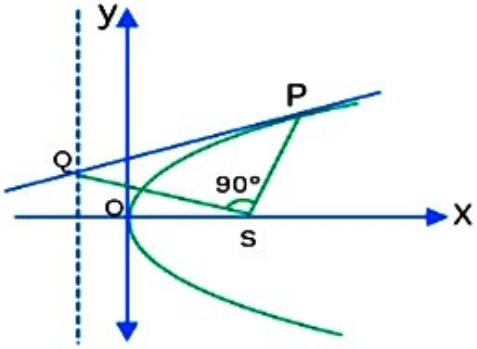
Slope at $SP = \frac{2at - 0}{at^2 - a} = \frac{2t}{t^2 - 1} = m_1$

Slope at $SQ = \frac{\frac{at^2 - a}{t} - 0}{-a - 0} = \frac{t^2 - 1}{-2t} = m_2$.

$\Rightarrow \quad m_1 m_2 = -1$

$\Rightarrow \quad SP \perp SQ$

$\Rightarrow \quad \angle PSQ = 90°$



(6) Tangent at P is $yt = x + at^2$ ... (i)

Line perpendicular to above line is $xt + y = \lambda$ and passes through (a,0) gives $\lambda = at$

$\therefore$ perpendicular line will be

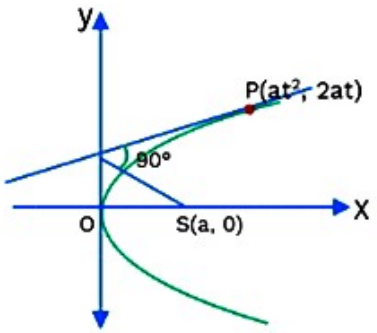
$xt + y = at$ ... (ii)

Solve (i) and (ii), we get

$x = 0$

i.e., these two lines intersect at y-axis i.e. tangent at the vertex.

The foot of the perpendicular from the focus on any tangent to a parabola lies on the tangent at vertex.



(7) Tangents and Normals at the extremities of the latus rectum of a parabola $y^2 = 4ax$ constitute a square, their points of intersection being (-a,0) & (3a,0).

(8) The circle circumscribing the triangle formed by any three tangents to a parabola passes through the focus.

(9) The orthocentre of any triangle formed by three tangents to a parabola $y^2=4ax$ lies on the directrix & has the co-ordinates $(-a, a(t_1+t_2+t_3+t_1 t_2 t_3))$

(10) The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

Paragraph for Q1 and Q3 :

Let a tangent to parabola $y^2 = 4ax$ at point $P(at^2, 2at)$, $t \neq 0$ intersects its directrix at point Q. Let 'S' represents the focus of parabola $y^2 = 4ax$ and C represents the circle circumscribing the triangle PQS.

Q.1 **The angle between the parabola $y^2 = 4ax$ and the circle C at point P is :**

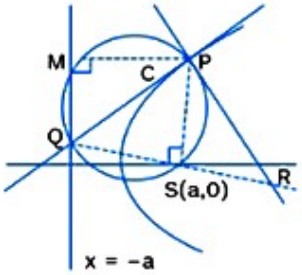
(A) $\frac{\pi}{6}$ **(B)** $\frac{\pi}{4}$ **(C)** $\frac{\pi}{3}$ **(D)** $\frac{\pi}{2}$

Ans. **(D)**

PQ is diameter of circle

$\Rightarrow$ tangent of parabola is normal to C

$\therefore$ Circle and parabola are orthogonal



Q.2 **If normal to parabola $y^2 = 4ax$ at point P intersects the line joining Q and S at R, then $\frac{(PS)(QR)}{(PQ)(PR)}$ is equal to :**

(A) 4 **(B) 3** **(C) 2** **(D) 1**

Ans. **(D)**

Area of $\Delta PQR = \frac{1}{2}(PS)(QR) = \frac{1}{2}(PQ)(PR) \Rightarrow \frac{(PS)(QR)}{(PQ)(PR)} = 1$

Q.3 **Area of circle C is:**

(A) $\frac{\pi a^2\left(1+t^2\right)^3}{8t^2}$ **(B)** $\frac{\pi a^2\left(1+t^2\right)^3}{4t^4}$ **(C)** $\frac{\pi a^2\left(1+t^2\right)^3}{4t^2}$ **(D)** $\frac{\pi a^2\left(1-t^2\right)^3}{4t^2}$

Sol. **(C)**

$$(PQ)^2 = \left(at^2+a\right)^2 + \left(2at - \frac{at^2-a}{t}\right)^2$$

$$= a^2\left[\left(t^2+1\right)^2 + \frac{\left(t^2+1\right)^2}{t^2}\right] = \frac{\left(t^2+1\right)^3 a^2}{t^2}$$

$$\text{Area of C} = \frac{\pi}{4}(PQ)^2 = \frac{\pi}{4}\frac{\left(t^2+1\right)^3}{t^2}a^2$$

Q. **Two mutually perpendicular tangent of the parabola $y^2 = 4ax$ meet the axis in P_1 and P_2. If S is the focus of the parabola, then $\frac{1}{(SP_1)} + \frac{1}{(SP_2)}$ is equal to**

(A) $\frac{4}{a}$ **(B)** $\frac{2}{a}$ **(C)** $\frac{1}{a}$ **(D)** $\frac{1}{4a}$

Sol. **(C)**

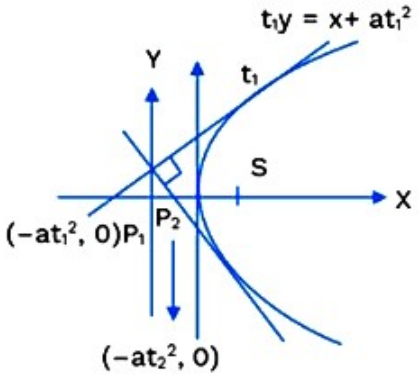
$SP_1 = a\left(1+t_1^2\right); SP_2 = a\left(1+t_2^2\right)$

$\Rightarrow \quad t_1t_2 = -1$

$$\frac{1}{SP_1} = \frac{1}{a\left(1+t_1^2\right)}$$

$$\frac{1}{SP_2} = \frac{1}{a\left(1+t_2^2\right)} = \frac{t_1^2}{a\left(t_1^2+1\right)}$$

$$\therefore \frac{1}{SP_1} + \frac{1}{SP_2} = \frac{1}{a}$$



Q. **A line of slope $\lambda(0 < \lambda < 1)$ touches parabola $y + 3x^2 = 0$ at P. If S is the focus and M is the foot of the perpendicular of directrix from P, then tan ∠MPS equals**

(A) 2λ **(B)** $\frac{2\lambda}{-1+\lambda^2}$ **(C)** $\frac{1-\lambda^2}{1+\lambda^2}$ **(D) None of these**

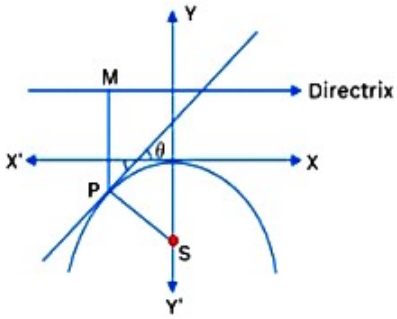
Sol. **(B)**

Slope of line

$\lambda = \tan\theta$

$$\tan(\angle MPS) = \tan 2\left(\frac{\pi}{2} - \theta\right) = \tan(\pi - 2\theta) = -\tan 2\theta$$

$$= \frac{2\lambda}{\lambda^2 - 1}$$



Q. **If the locus of middle of point of contact of tangent drawn to the parabola $y^2 = 8x$ and foot of perpendicular drawn from its focus to the tangents is a conic then length of latus rectum of this conic is:**

(A) $\frac{9}{4}$ **(B) 9** **(C) 18** **(D)** $\frac{9}{2}$

Sol. **(B)**

Let middle point of P and T be (h, k)

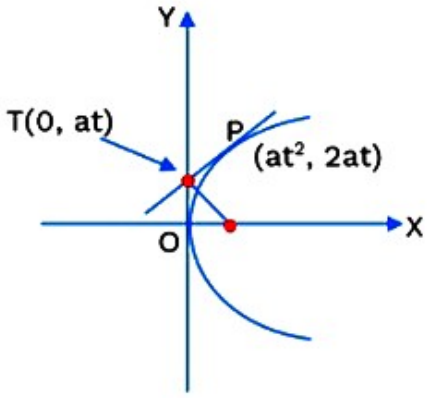
$\therefore \quad 2h = at^2$

and $2k = 3at$

$\therefore \quad 2h = a\dfrac{4k^2}{9a^2}$

Locus of (h, k) is $2y^2 = 9ax$

As $a = 2 \qquad \therefore y^2 = 9x$



Q. **From a point A(t) on the parabola $y^2 = 4ax$, a focal chord and a tangent is drawn. Two circles are drawn in which one circle is drawn taking focal chord AB as diameter and other is drawn by taking intercept of tangent between point A and point P on the directrix, as diameter. Then the common chord of the circles is**

(A) line joining focus and P

(B) line joining focus and A

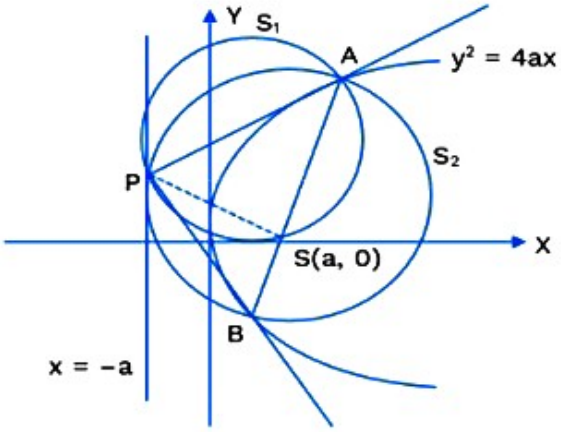
(C) tangent to the parabola at point A

(D) None of these

Sol. **(C)**

Circle S_2, taking focal chord AB as diameter will touch directrix at point P and circle S_1, taking AP as diameter will pass through focus S(since AP subtends angle 90° at focus of parabola).

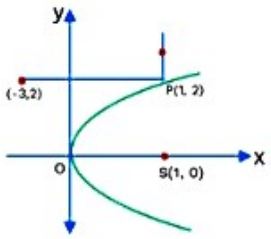
Hence, common chord of given circles is line AP (which is intercept of tangent at point 'A' between point A and directrix).



Q. **If incident ray from point (-3,2) parallel to the axis of parabola $y^2=4x$ strike the parabola, then find the equation of reflected ray.**

Sol. Since incident ray strikes parabola at P(1,2) i.e. extremity of latus rectum

and it will pass through the focus of parabola therefore reflected ray will be parallel to y -axis and its equation will be x = 1.



Q. **A ray of light moving parallel to the x -axis get reflected from a parabolic mirror $(y-2)^2=4(x+1)$. Find the point on the axis of parabola through which the ray must pass after reflection.**

Sol. Axis of parabola is y = 2 i.e., parallel to x -axis. As we know if incident ray is parallel to x -axis then after reflection it will pass through the focus of parabola and focus is (0, 2).

Paragraph for Q1 to Q3 :

Two tangents on a parabola are x – y = 0 and x + y = 0 . S(2, 3) is the focus of the parabola.

Q1 **The equation of tangent at vertex is**

(A) 4x – 6y + 5 = 0 **(B) 4x – 6y + 3 = 0**

(C) 4x – 6y + 1 = 0 **(D) 4x – 6y + 3/2 = 0**

Ans. **(A)**

Q2 **The length of latus rectum of the parabola is**

(A) $6/\sqrt{3}$ **(B)** $10/\sqrt{13}$ **(C)** $2/\sqrt{13}$ **(D) None of these**

Ans. **(B)**

Q3 **If P and Q are the ends of the focal chord of the parabola, then** $\frac{1}{SP}+\frac{1}{SQ}=$

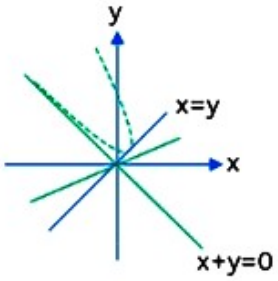
(A) $2\sqrt{13}/3$ **(B)** $2\sqrt{13}$ **(C)** $2\sqrt{13}/5$ **(D) None of these**

Sol. **(C)**

We know that the foot of perpendicular from focus to any tangent lies on the tangent at the vertex of the parabola. Now, if the foot of perpendicular of (2, 3) on the line $x-y=0$ is (x_1, y_1), then

$$\frac{x_1-2}{1}=\frac{y_1-3}{-1}=\frac{-(2-3)}{2}$$

or $x_1=\frac{5}{2}$ and $y_1=\frac{5}{2}$



If the foot of perpendicular of (2, 3) on the line $x+y=0$ is (x_2,y_2), then

$$\frac{x_2-2}{1}=\frac{y_2-3}{1}=-\frac{2+3}{2}$$

or $x_2=-\frac{1}{2}$ and $y_2=\frac{1}{2}$

Now, the tangent at the vertex passes through the points (5 /2, 5/2) and $(-1/2, 1/2)$. Then, its equation is

or $y-\frac{1}{2}=\frac{2}{3}\left(x+\frac{1}{2}\right)$

or $4x-6y+5=0$

The length of latus rectum of the parabola is 4 × (Distance of focus from tangent at vertex)

$$=4\times\left|\frac{8-18+5}{\sqrt{52}}\right|=\frac{10}{\sqrt{13}}$$

Also, the distance between the focus and the tangent at vertex is $5/\sqrt{13}$

We know that $\frac{1}{SP}+\frac{1}{SQ}=\frac{1}{a}$

where a is 1/4 th of the length of latus rectum. Therefore,

$$\frac{1}{SP}+\frac{1}{SQ}=\frac{2\sqrt{13}}{5}$$

Q. **Let S be the focus of the parabola $y^2=8x$ and PQ be the common chord of the circle $x^2+y^2-2x-4y=0$ and the given parabola. The area of triangle PQS is**

Sol. **(4)**

We have parabola $y^2=8x$ with circle

$x^2+y^2-2x-4y=0$

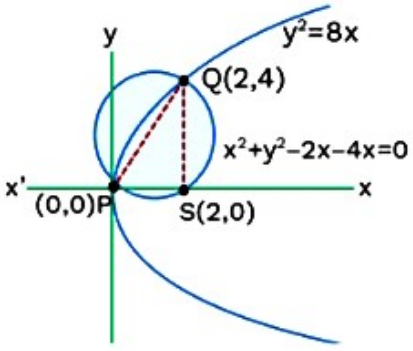
Solving parabola $(2t^2, 4t)$ with circle,

we get $4t^4+16t^2-4t^2-16t=0$

or $t^4+3t^2-4t=0$

$\Rightarrow t=0, 1$

So, the points P and Q are (0,0) and (2, 4),



respectively which are also diametrically opposite points on the circle. The focus is S(2, 0)

Area of $\Delta PQS = \frac{1}{2} \times 2 \times 4 = 4$ sq. units

Q. **Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is**

Sol. **(4)**

Let $P(t^2, 2t)$ be a point on the curve $y^2 = 4x$ and Q(h, k) be it's image in $x + y + 4 = 0$.

$$\therefore \quad \frac{h - t^2}{1} = \frac{k - 2t}{1} = -\frac{2\left(t^2 + 2t + 4\right)}{2}$$

$\Rightarrow h = -(2t + 4)$ and $k = -(t^2 + 4)$

Now $y = -5$ intersect this locus.

$\therefore k = -5$, So $t = \pm 1$, so $t = \pm 1$

Hence, $h = -2, -6$

So, points of intersection are A(–2, –5) and B(–6, – 5).

Hence, AB = 4.